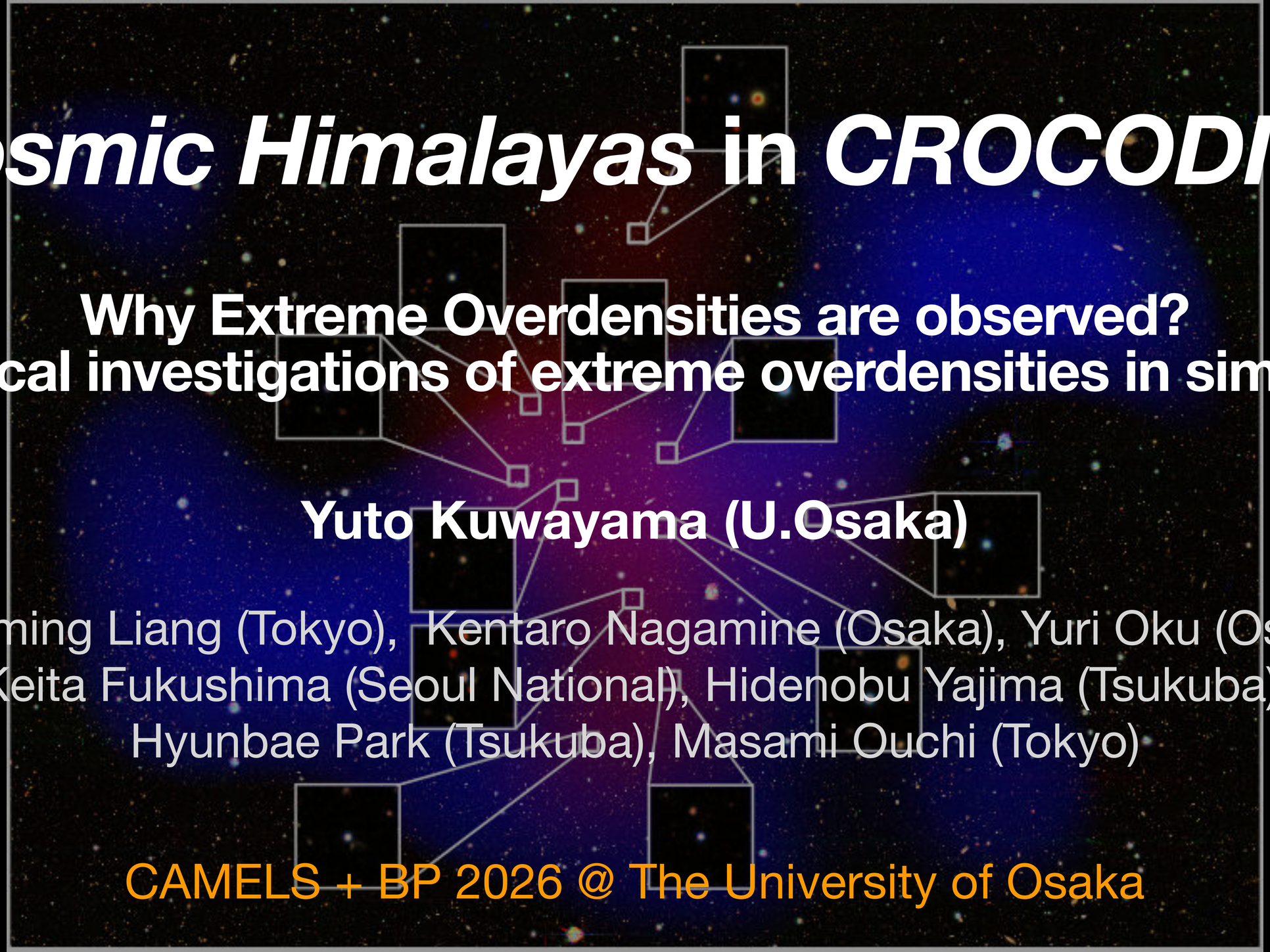


Cosmic Himalayas in *CROCODILE*

Why Extreme Overdensities are observed?
Statistical investigations of extreme overdensities in simulation

Yuto Kuwayama (U.Osaka)

Yongming Liang (Tokyo), Kentaro Nagamine (Osaka), Yuri Oku (Osaka),
Keita Fukushima (Seoul National), Hidenobu Yajima (Tsukuba),
Hyunbae Park (Tsukuba), Masami Ouchi (Tokyo)

CAMELS + BP 2026 @ The University of Osaka

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Cosmic Himalayas in CROCODILE : Probing the Extreme Quasar Overdensities by Count-in-Cells analysis and Nearest Neighbor Distribution

Kuwayama, Yuto ; Liang, Yongming ; Nagamine, Kentaro ; Oku, Yuri ; Nishihama, Daisuke ; Toyouchi, Daisuke ; Fukushima, Keita ; Yajima, Hidenobu ; Park, Hyunbae ; Ouchi, Masami *show details*

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The recently reported Cosmic Himalayas (CH) -- an extreme quasar overdensity at z~2 -- poses an apparent challenge to the Lambda CDM framework, with a reported significance of 16.9-sigma under Gaussian assumptions. Such an event appears improbably rare, with a formal probability of P ~ 10^-68. In this work, we investigate whether CH-like structures can naturally arise in cosmological hydrodynamic simulations. Using the CROCODILE simulation, which self-consistently models galaxy-black hole coevolution, we examine quasar clustering through two complementary approaches: the count-in-cells (CIC) statistic, which probes large-scale overdensities, and the nearest-neighbor distribution (NND), sensitive to small-scale environments. CIC analysis reveals that the underlying distribution is heavy-tailed and non-Gaussian, and that conventional Gaussian-based evaluation substantially overestimates the significance of extreme events. When modeled with an asymmetric generalized normal distribution (AGND), the inferred rarity of the CH is substantially reduced and reconciled with standard Lambda CDM; for instance, regions appearing as 12-sigma outliers under Gaussian assumptions (P ~ 10^-33) are found to occur in the AGND regime with a probability of P ~ 10^-4. NND analysis further demonstrates that extreme overdense regions within the simulation can naturally sustain two-point correlation function values similar to those observed in the CH (r0 ~ 30 Mpc/h), suggesting that the strong clustering stems from sample selection biases and local environmental variations. These two analyses conclusively highlight the importance of adopting non-Gaussian statistics when quantifying extreme overdensities of quasars and establish that the CH is not an anomaly, but a natural outcome of structure formation in the Lambda CDM universe.

Publication eprint arXiv:2512.24966

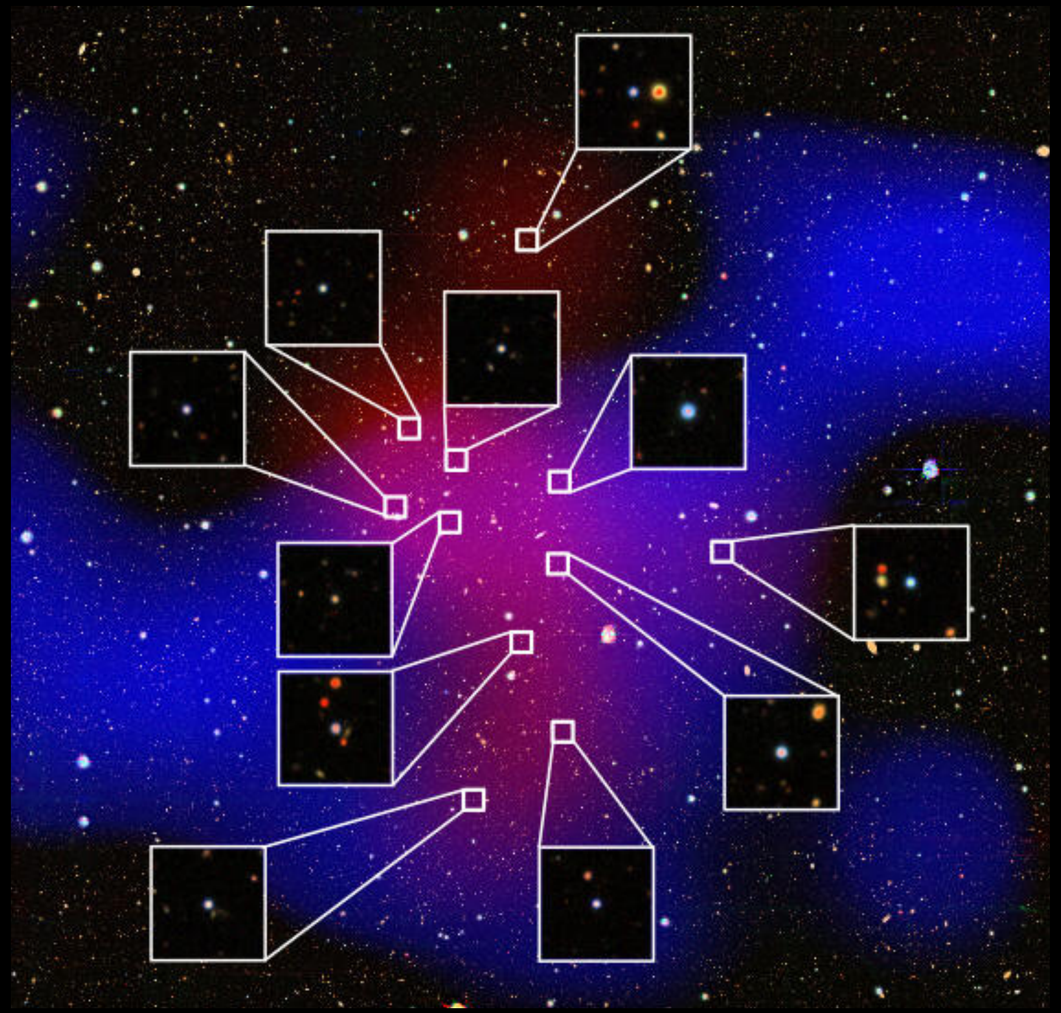
SciX link

YK+2025 (Accepted in ApJ on March) arXiv: 2512.24966

“*The Cosmic Himalayas*” (CH): The Extremely Overdensity of Quasars

Liang+2025 (SDSS, rest-frame UV)

c.f. Cai+2017, Mukae+2020, Shi+2021



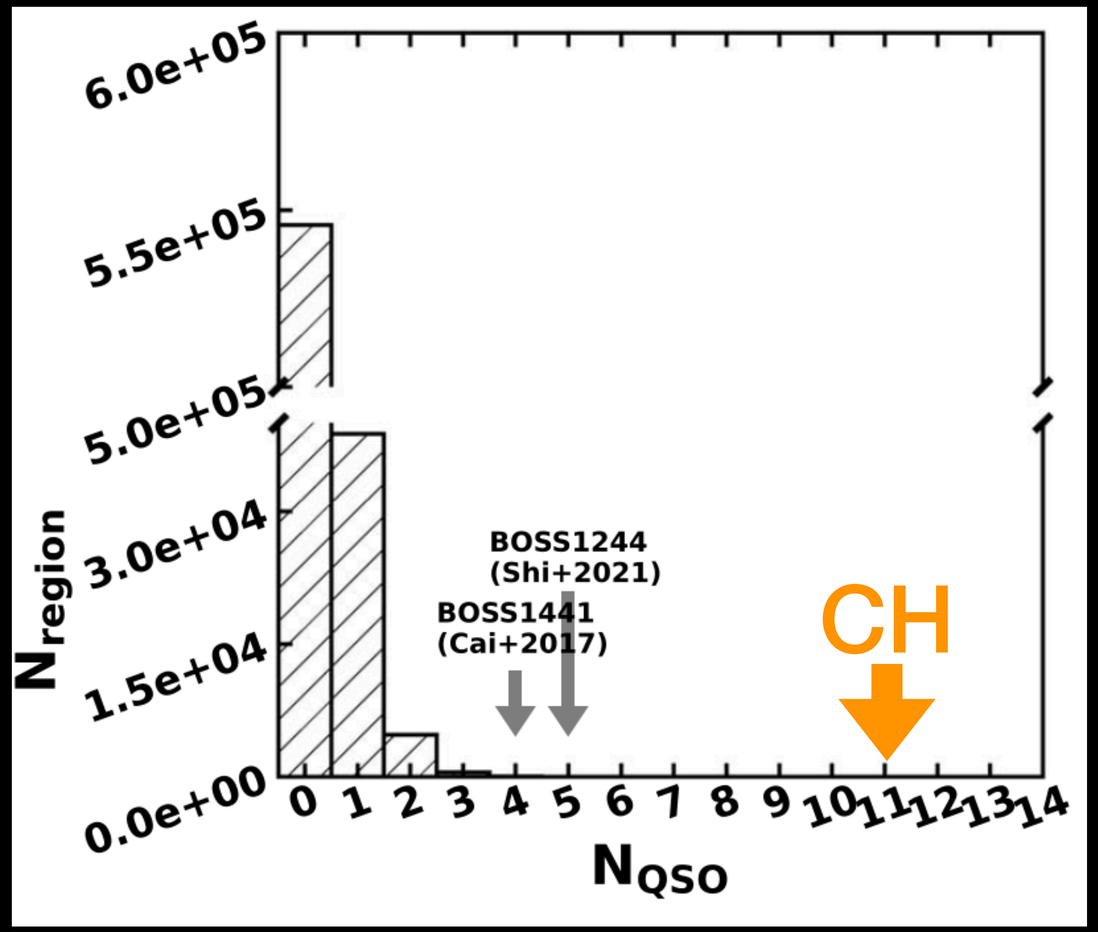
11 QSOs in $L \simeq 30\,h^{-1}\,\mathrm{Mpc}$ box

($\delta_{\mathrm{QSO}} = 30$, $z = 2.16 - 2.20$)

© Subaru Telescope

"*The Cosmic Himalayas*" (CH): The Extremely Overdensity of Quasars

Liang+2025 (SDSS, rest-frame UV)



11 QSOs in $L \simeq 30\,h^{-1}\,\mathrm{Mpc}$ box

($\delta_{\mathrm{QSO}} = 30$, $z = 2.16 - 2.20$)

Assuming Gaussian Shape CIC

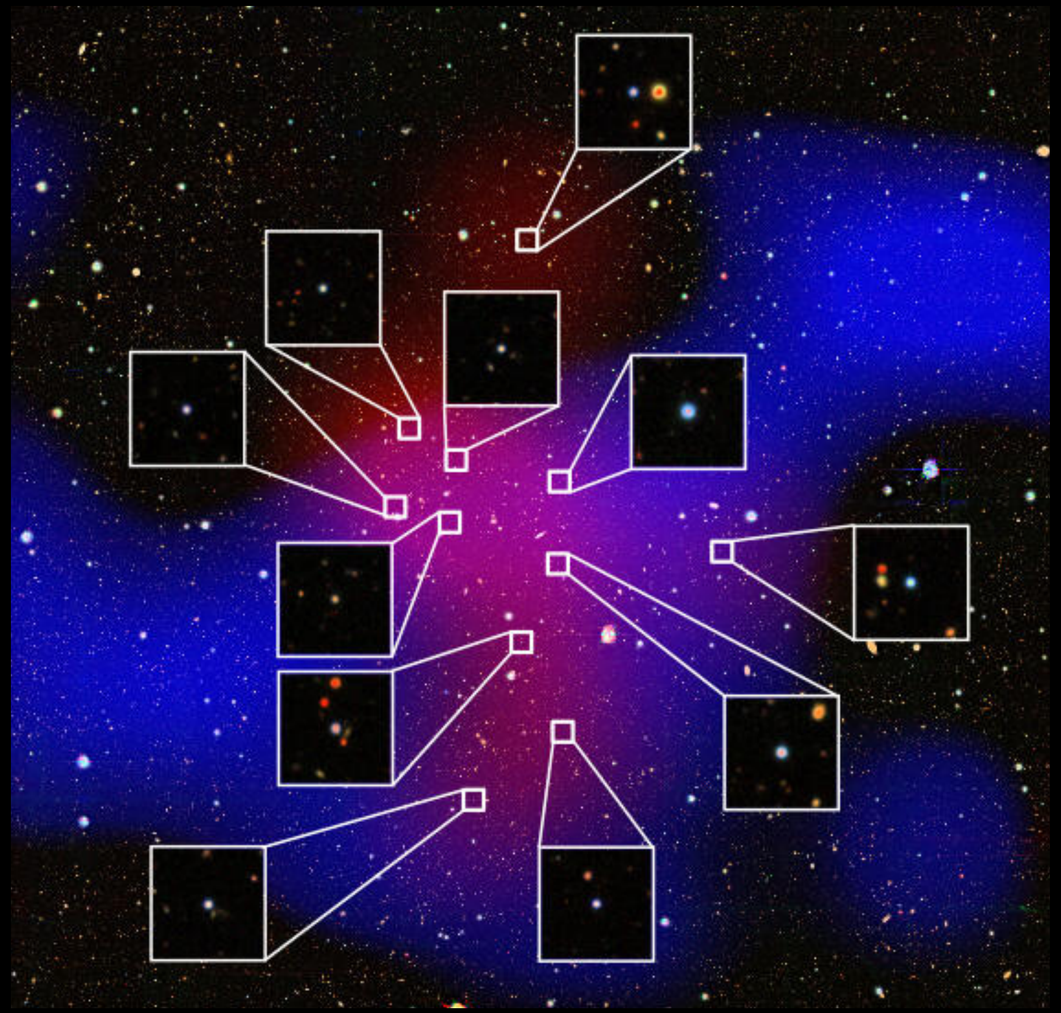
$\rightarrow\ \delta_{\mathrm{QSO}} = 16.9\sigma$ ($P \sim 10^{-68}$)

Count-in-cells analysis

(e.g. Ueda & Yokoyama 1996, Shin+2017)

“*The Cosmic Himalayas*” (CH): The Extremely Overdensity of Quasars

Liang+2025 (SDSS, rest-frame UV)



© Subaru Telescope

11 QSOs in $L \simeq 30\,h^{-1}\,\mathrm{Mpc}$ box

($\delta_{\mathrm{QSO}} = 30$, $z = 2.16 - 2.20$)

$\rightarrow\ \delta_{\mathrm{QSO}} = 16.9\sigma$ ($P \sim 10^{-68}$)

Questions

- Can we find **similar regions with CH** from simulations?
- **Why** are CH **so overdense**?

Outline: Investigating the extreme quasar overdensities

Statistically searching for extreme quasar overdensities analogous to CH

Cosmological hydrodynamic simulation CROCODILE at $z = 2.2$
(Oku & Nagamine 2024)

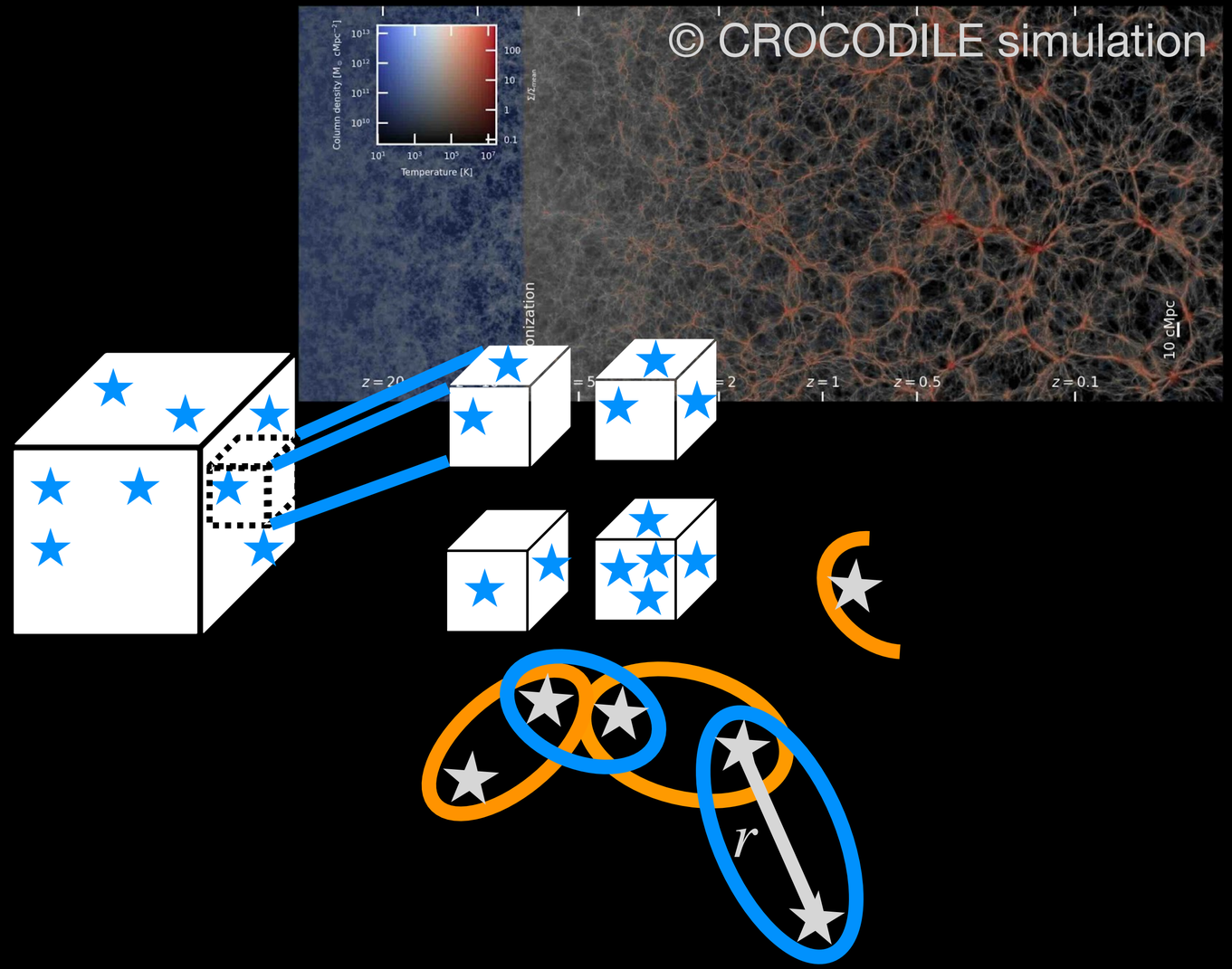
-> Select Quasar sample

+

Two Complementary Statistics

- Count-in-cells (CIC) analysis

- Nearest Neighbour Distribution (NND)



Outline: Investigating the extreme quasar overdensities

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Cosmological hydrodynamic simulation CROCODILE at $z = 2.2$

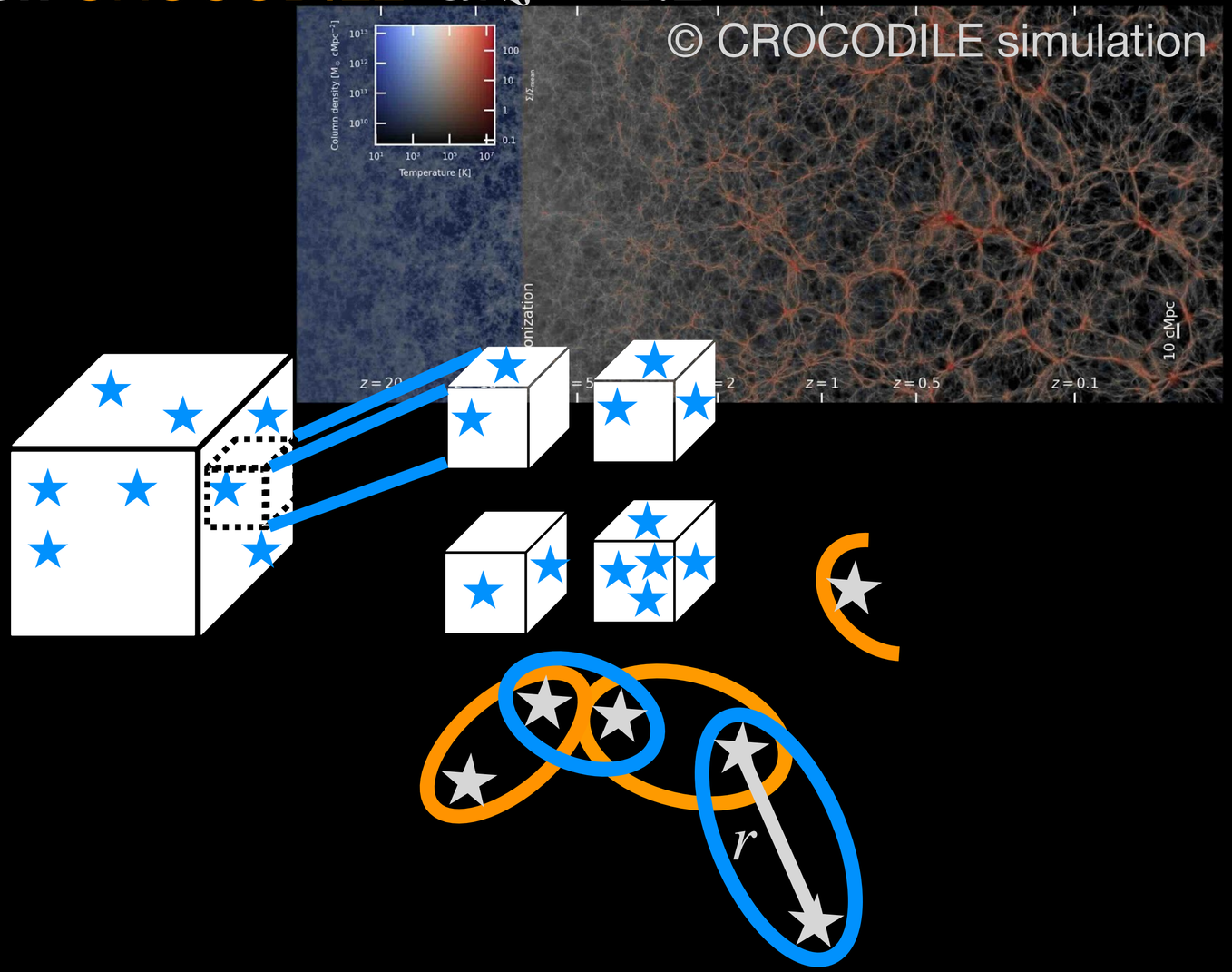
-> Select Quasar sample

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Two Complementary Statistics

- Count-in-cells (CIC) analysis

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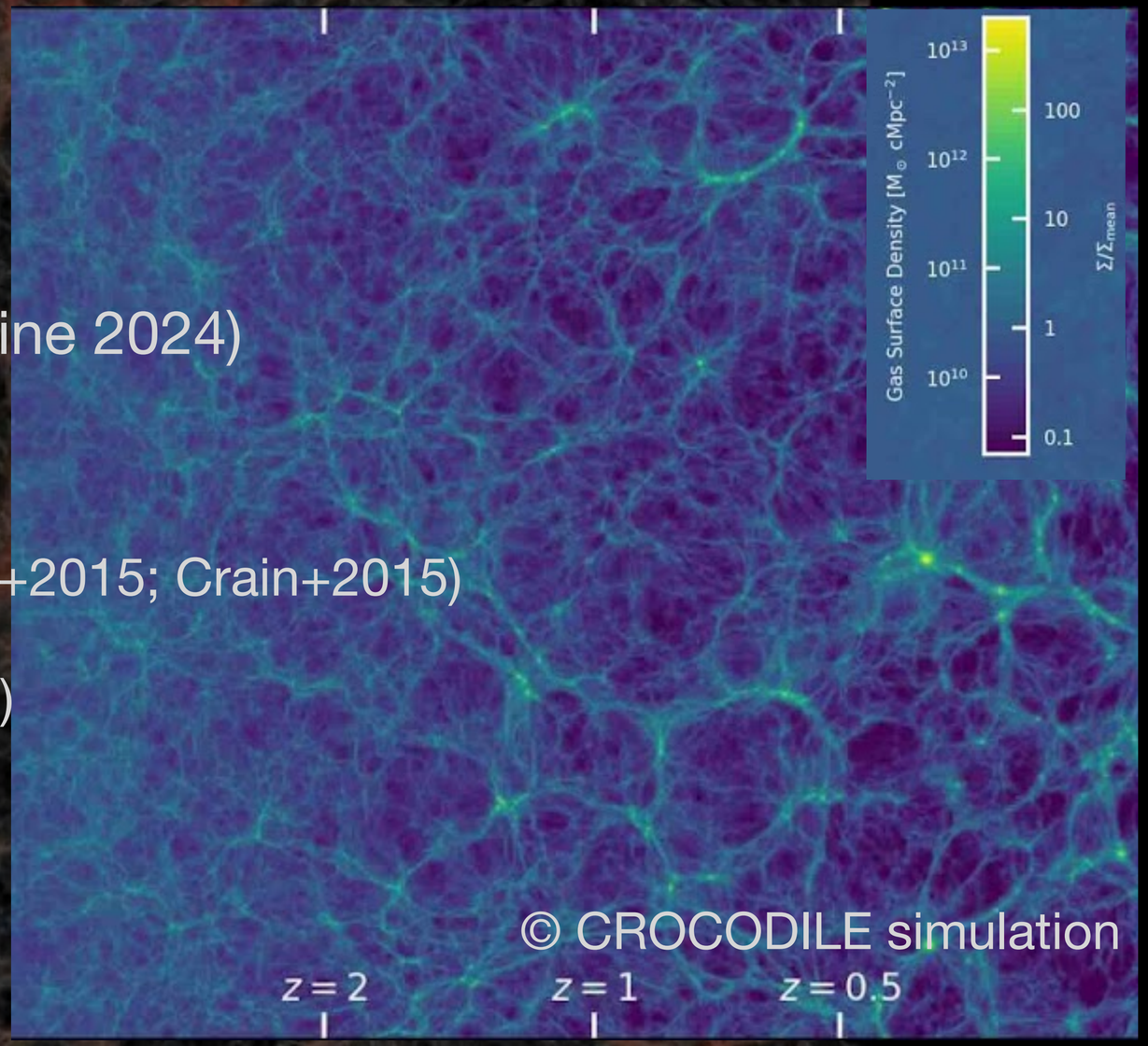
CROCODILE simulation (For making mock Quasar distribution)

CROCODILE (Oku & Nagamine 2024) version2 (YK+2025,Nishihama+ in prep.)

Cosmological hydrodynamic simulation
(e.g. Davé+2019; Hopkins+2022; Schaye+2025)

Code: GADGET4-Osaka SPH/N-body
(Romano+2022a,b; Oku+2022; Oku & Nagamine 2024)

- SN feedback (Oku & Nagamine 2024)
- EAGLE-like Thermal AGN feedback (Schaye+2015; Crain+2015)
- Chemical enrichment by CELib (Saitoh+2017)
- Cooling by GRACKLE (Smith+2017)
- etc…



BH population in CROCODILE at $z = 2.2$ (Consistent with obs)

Estimating Luminosity

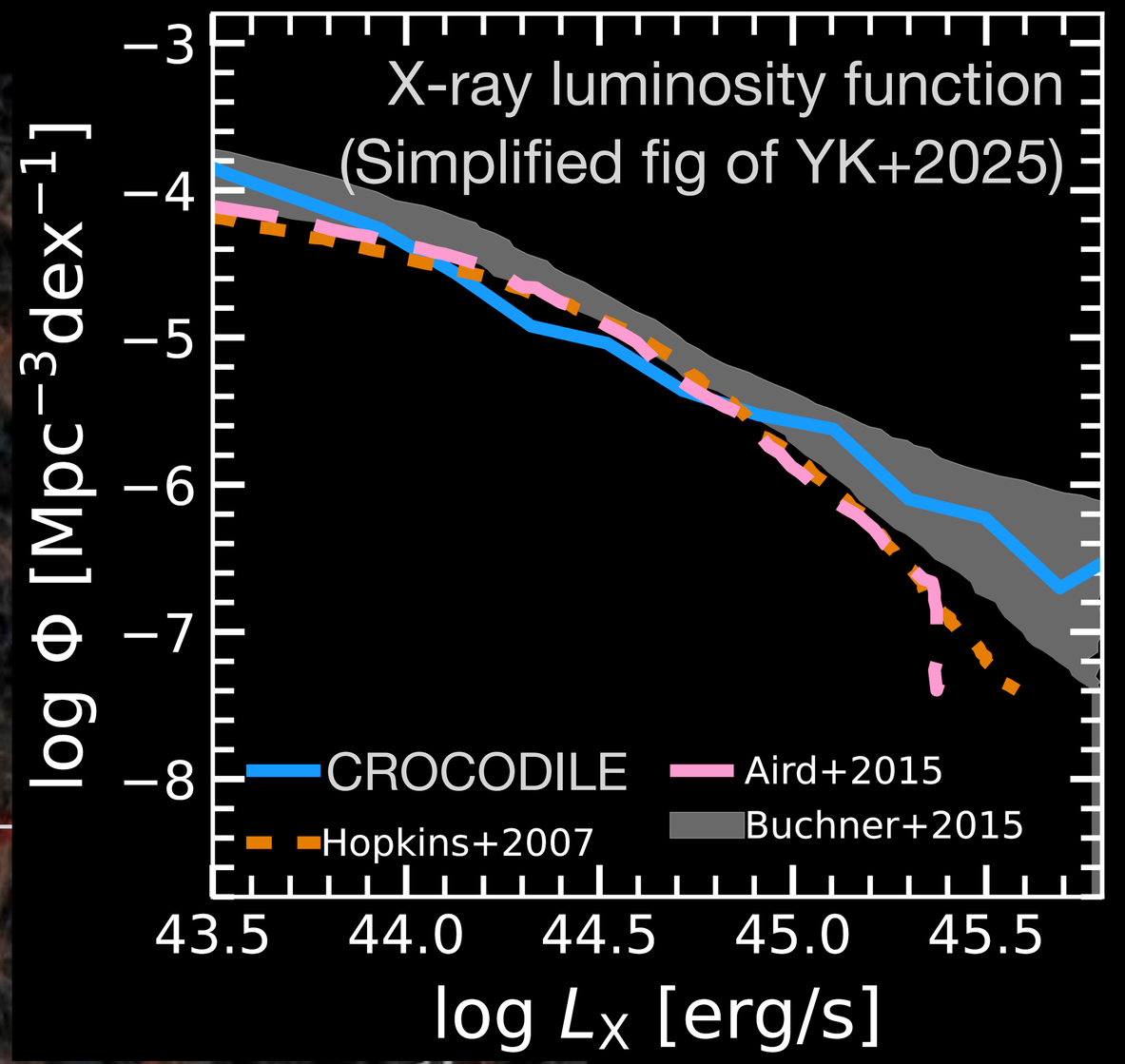
$\dot{M}_{\rm acc} = \alpha C \dot{M}_{\rm Bondi}$ (CROCODILE gives)

$L_{\rm bol} = \eta \dot{M}_{\rm acc} c^2 \qquad (\eta = 0.1)$

$\log(L_{\rm X}/L_\odot) = \log(L_{\rm bol}/L_\odot) - \log {\rm BC}$

$${\rm BC} = 10.83\left(\frac{L_{\rm bol}}{10^{10}L_\odot}\right)^{0.28} + 6.08\left(\frac{L_{\rm bol}}{10^{10}L_\odot}\right)^{-}$$

(Hopkins+2007)



c.f. Habouzit+2022

Quasar Sampling Conditions (Mocking observations)

AGN selection:

Mass: $8.0 < \log M_{\rm BH}/M_\odot < 9.5$ [SDSS main samples (1σ): Rakshit+2020]

Luminosity: $\log L_{\rm bol}/{\rm erg\ s^{-1}} \geq 45.47$ [CH quasar luminosity: Liang+2025]

Type I AGN selection:

Statistically assign "Type I" labels based on observational $f_{\rm TypeI}$ [Merloni+2014]

Generate 100 independent mock realisations to marginalise over sampling noise.

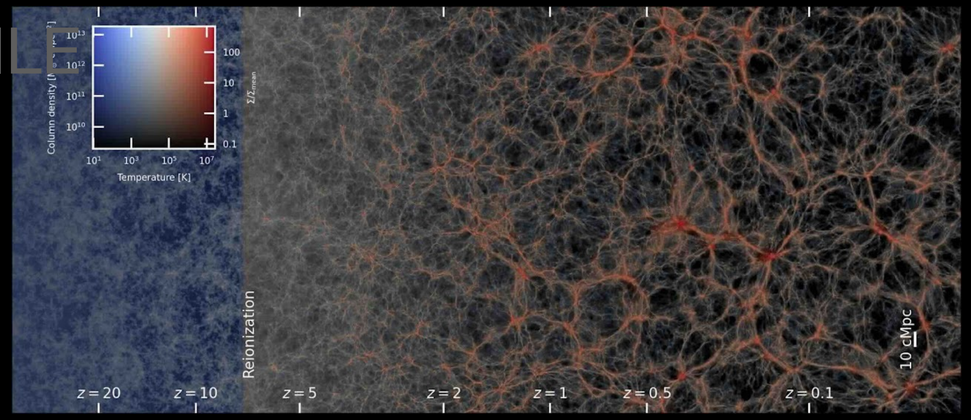
$$f_{\rm TypeI} = 0.44 - \frac{1}{\pi}\arctan\left(\frac{l_0 - l_x}{\sigma_x}\right) \text{ where } l_x = \log L_{\rm X}/{\rm erg\ s^{-1}},\ l_0 = 43.89,\ \sigma_x = 0.46$$

Outline: Investigating the extreme quasar overdensities

Statistically searching for extreme quasar overdensities analogous to CH

Cosmological hydrodynamic simulation CROCODIL

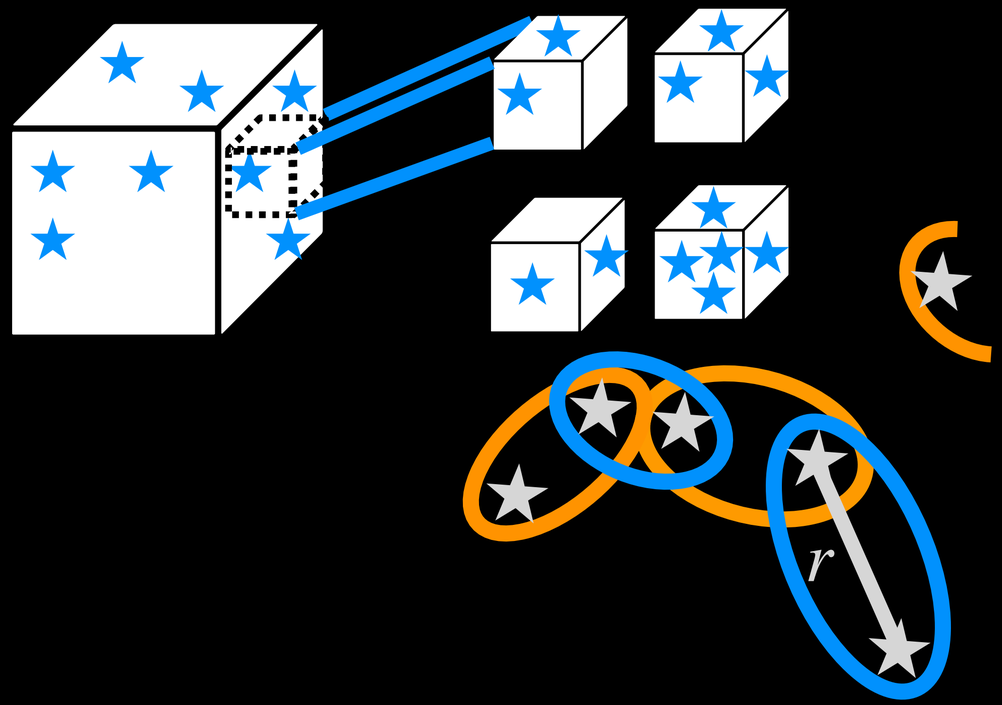
-> Select Quasar sample



+

Two Complementary Statistics

- Count-in-cells (CIC) analysis



- Nearest Neighbour Distribution (NND)

Clustering Evaluator : Count-in-cells (CIC)

1-pt Statistics measuring the global clustering
(e.g. Ueda & Yokoyama 1996, Shin+2017)

(Same analysis evaluating the significance of CH overdensity in Liang+2025)

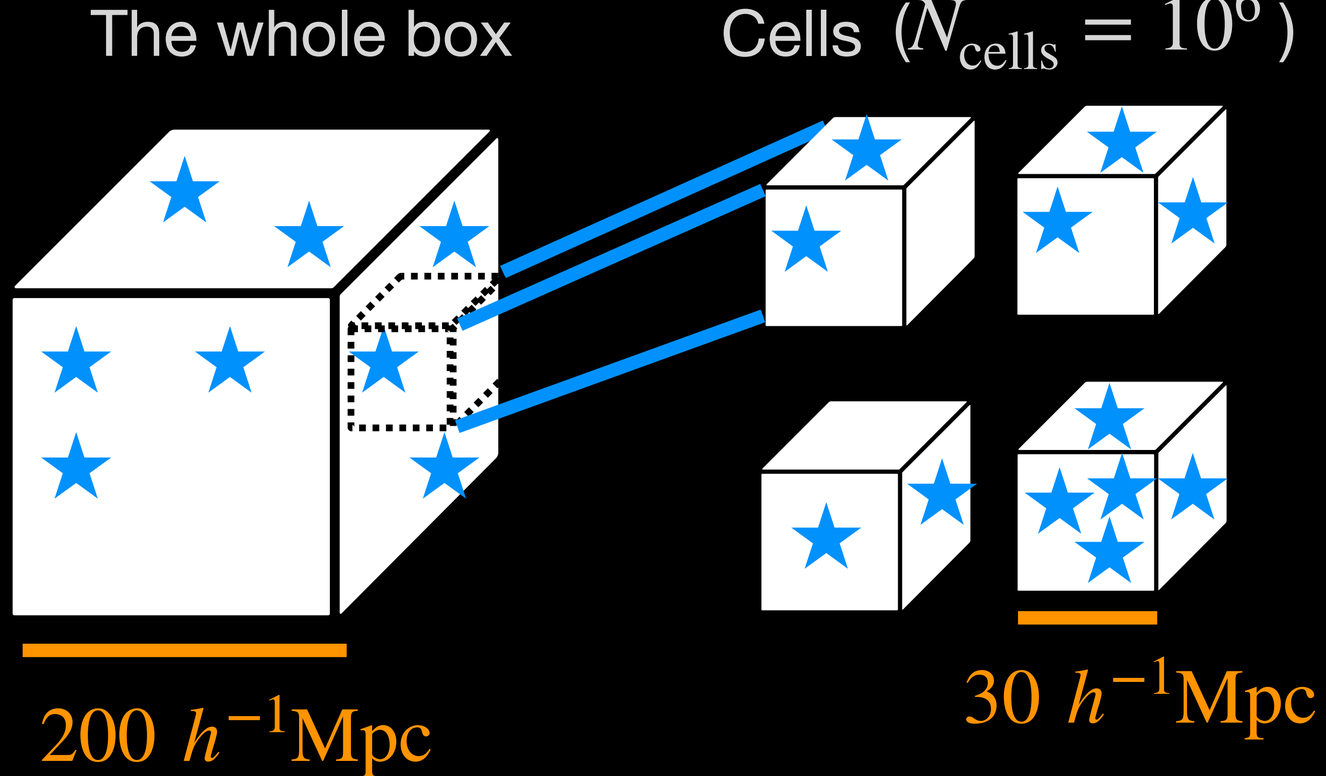
Divide volume into uniform cells

↓

Measure $N_{\rm QSO}$ in each cell

↓

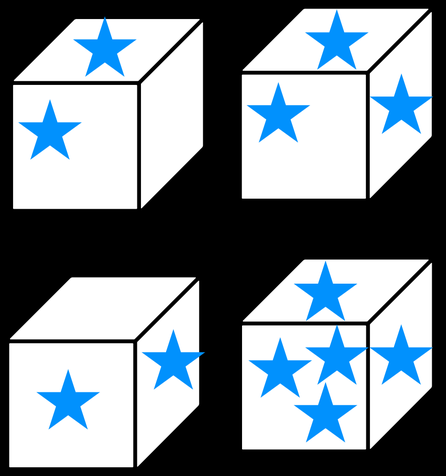
Construct probability distribution (PDF) $P_{\rm CIC}(N_{\rm QSO})$

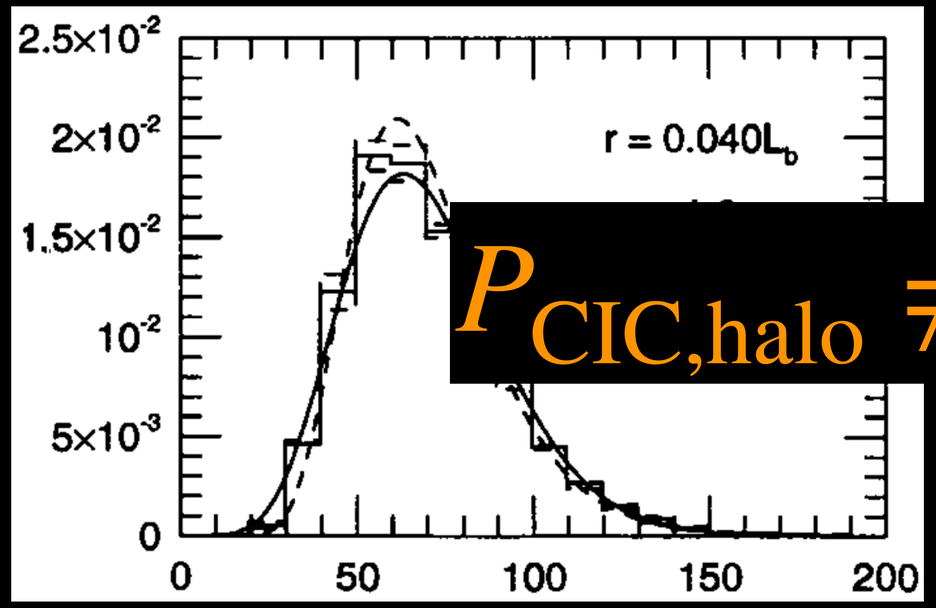


Counts-in-cells (CIC) analysis

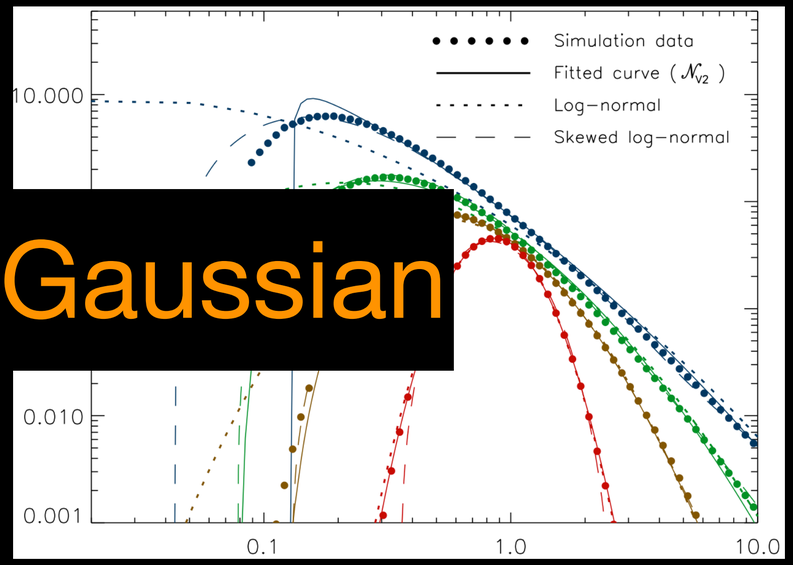
1-pt Statistics measuring the global clustering

CIC for dark matter haloes





$P_{\rm CIC,halo} \neq$ Gaussian



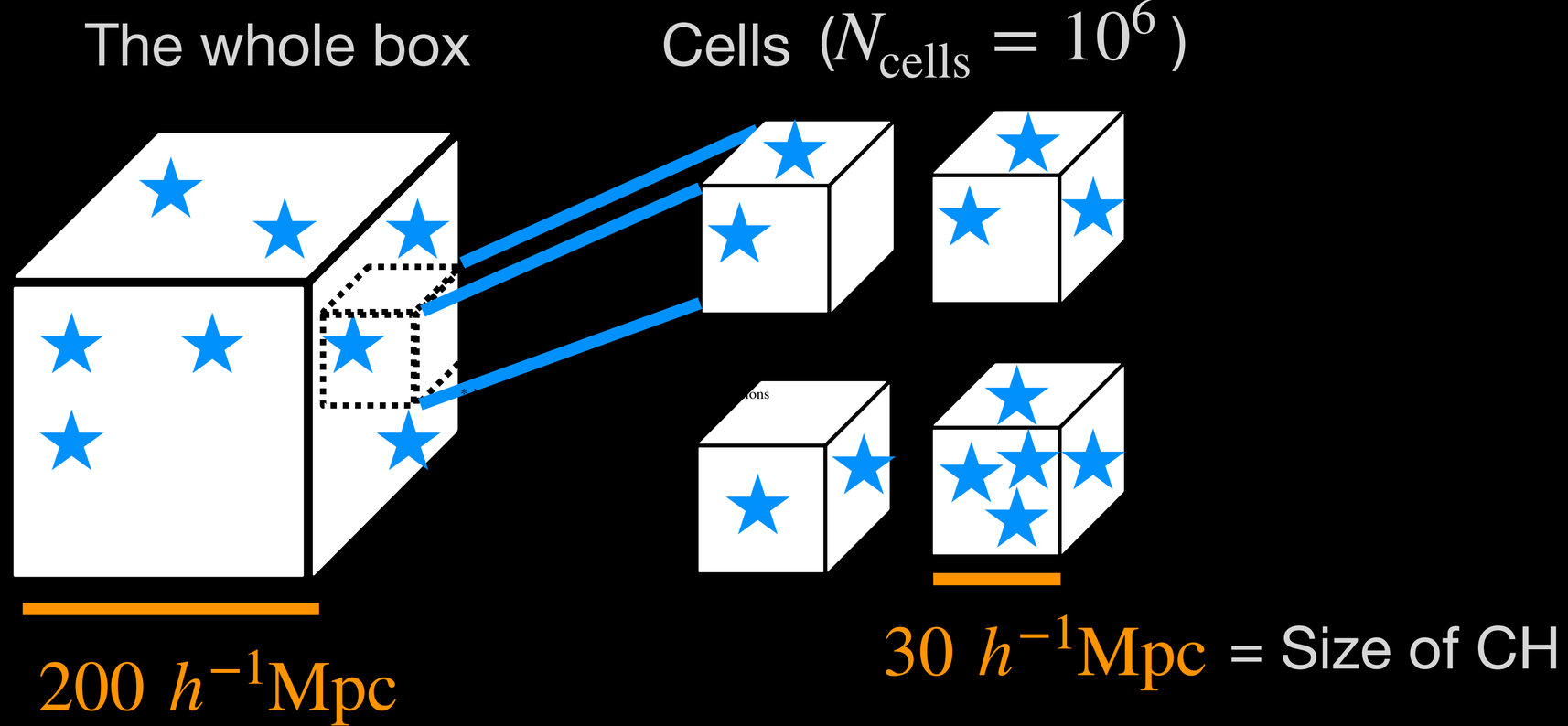
Ueda & Yokoyama 1996

CIC = log-normal shape

Shin+2017

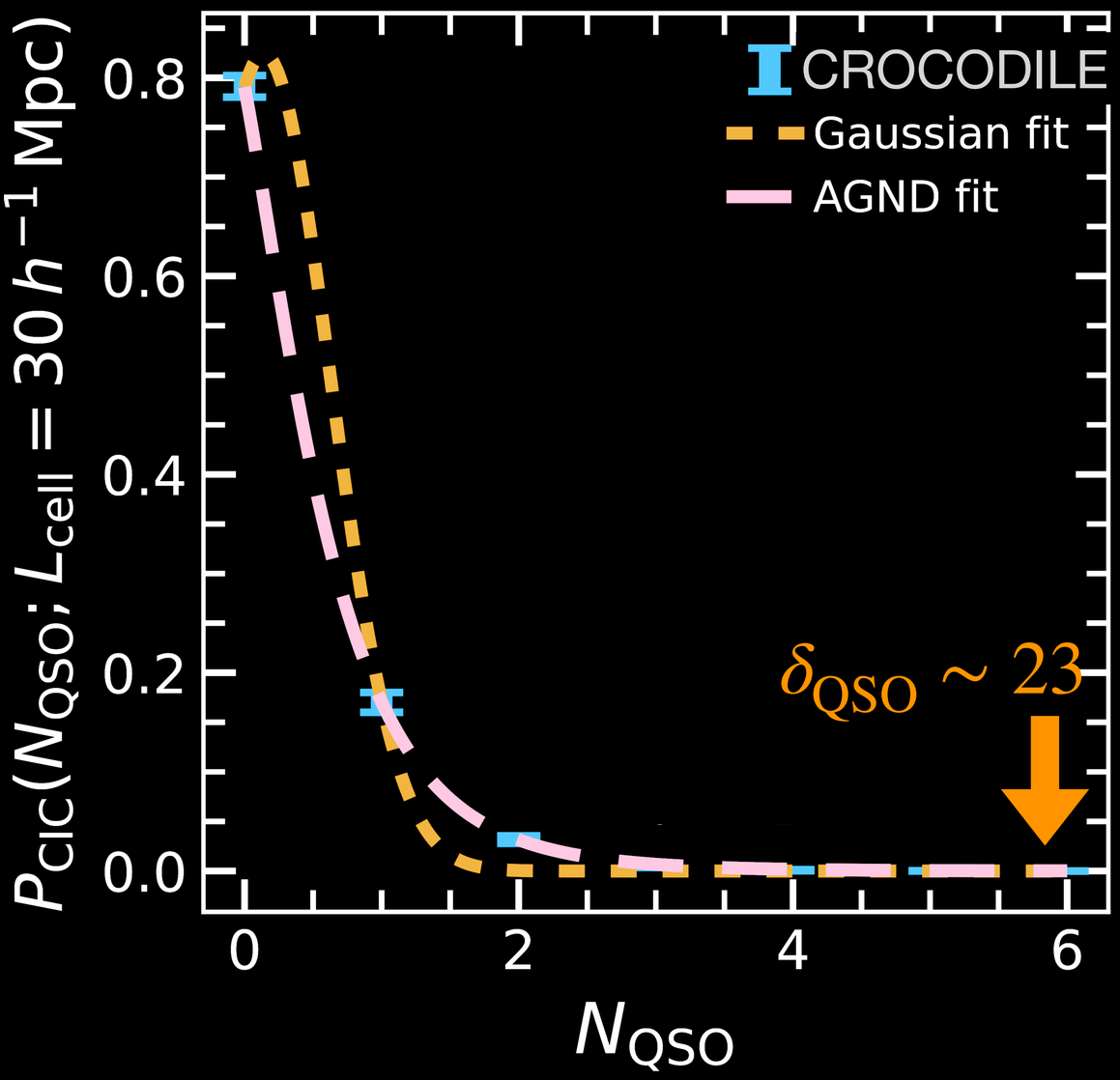
CIC = AGND shape

Setup: CIC analysis for CROCODILE

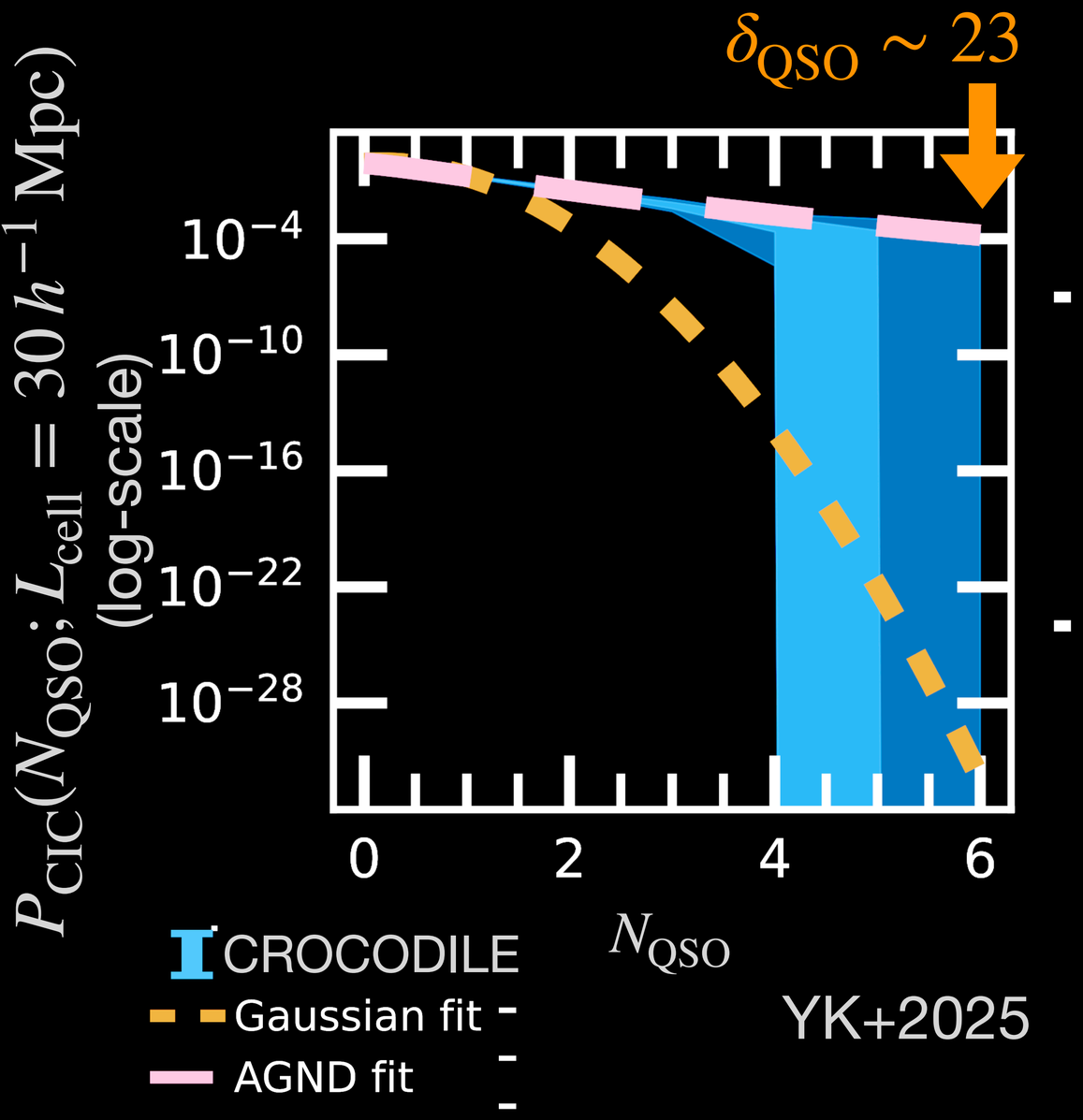


* 100 independent Type I full-box mock realisations
* Randomly selecting cells allowing overlapping cells

CIC result: Quasar CIC = Gaussian ?



CIC result: Quasar CIC shape ≠ Gaussian



✕ Gaussian

✓ Asymmetric generalised normal dist. (AGND)

(Likelihood Ratio Test: p value $\sim 10^{-29}$)

Assumption	Significance $\delta_{\rm QSO} \sim 23$	Conclusion
Gaussian	12σ ($P \sim 10^{-33}$)	Anomaly!
AGND	$P \sim 10^{-4}$	**Natural**

Outline: Investigating the extreme quasar overdensities

Statistically searching for extreme quasar overdensities analogous to CH

Cosmological hydrodynamic simulation CROCODILE

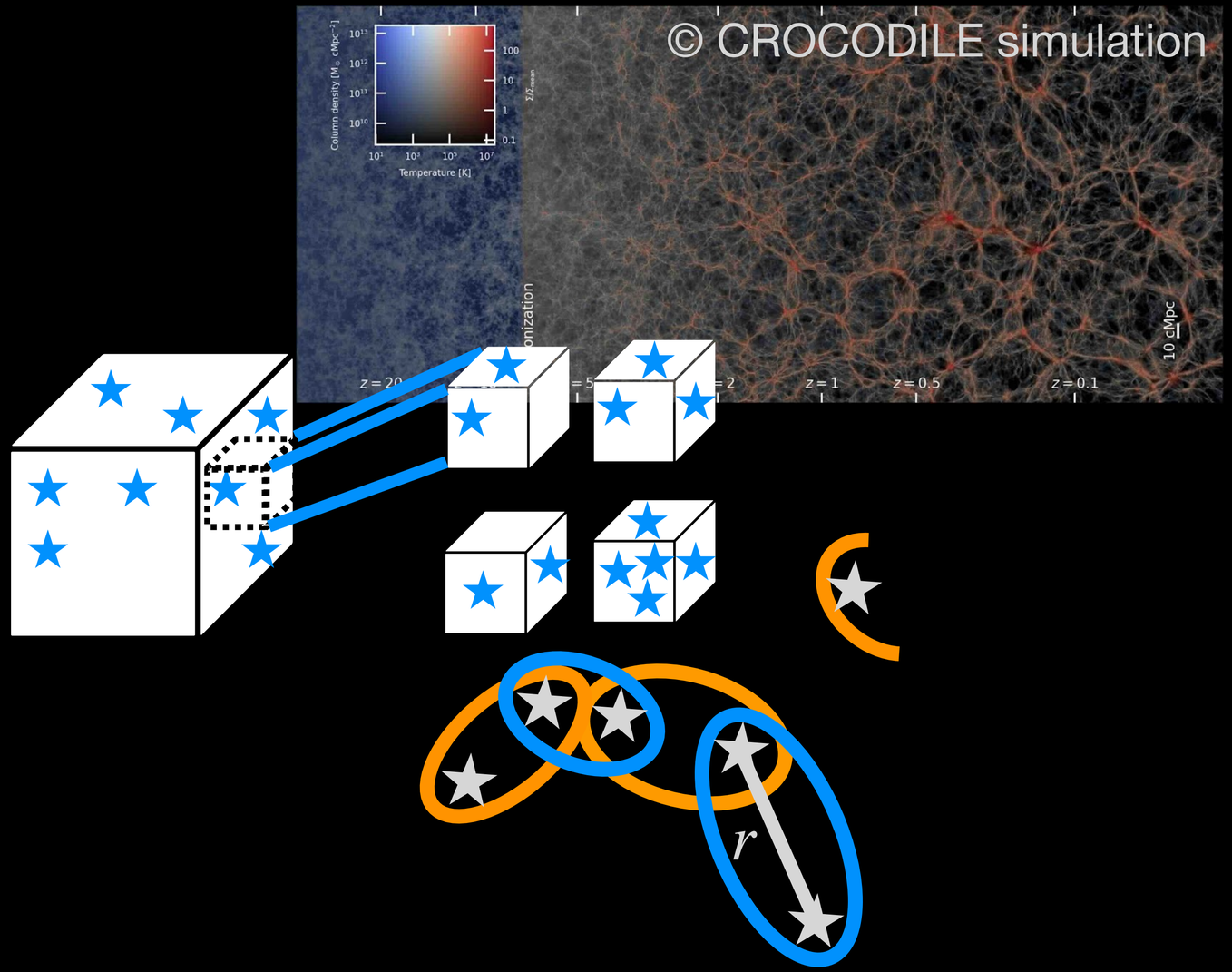
-> Select Quasar sample

+

Two Complementary Statistics

- Count-in-cells (CIC) analysis

- Nearest Neighbour Distribution (NND)

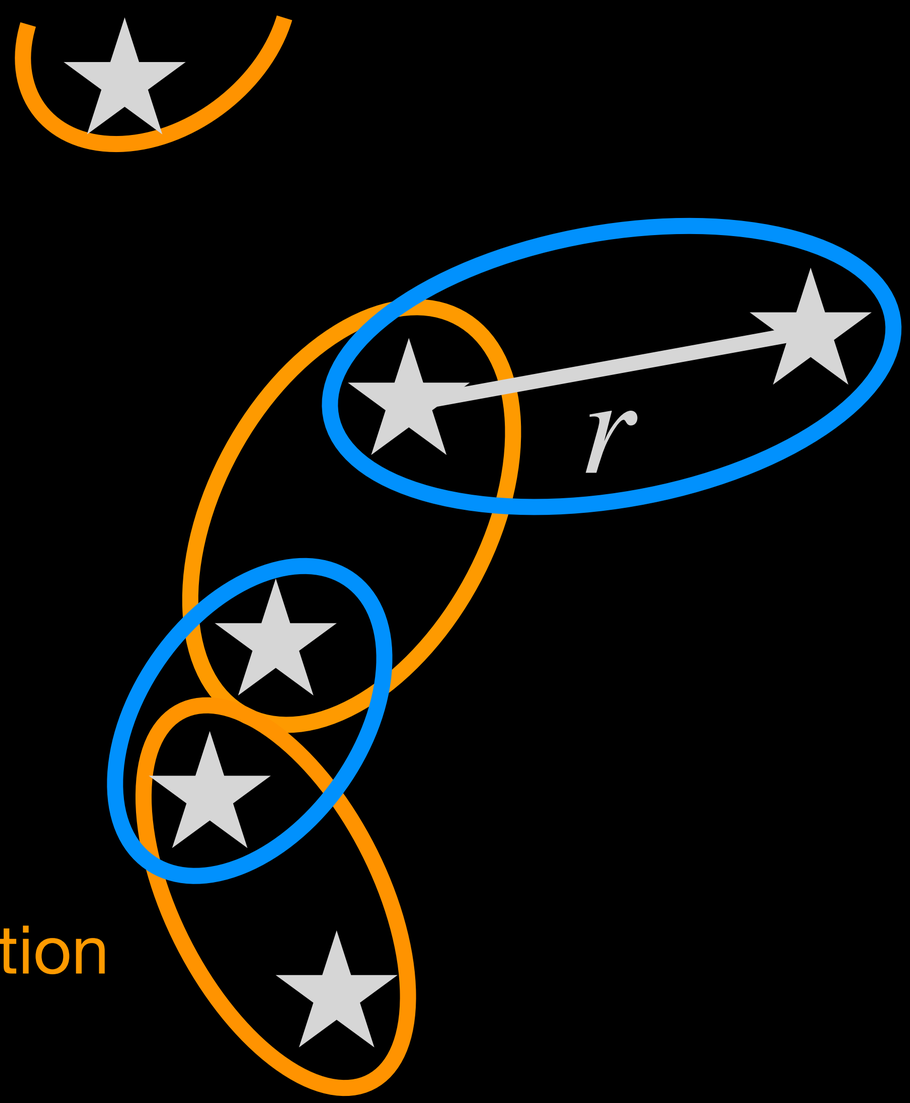


Nearest Neighbour Distribution (NND)

2-pt Statistics measuring local clustering (e.g. Banerjee & Abel 2021)

Probability Distribution of distances r

from an object to the nearest one $\frac{\mathrm{d}P_{\mathrm{NN}}}{\mathrm{d}r}(<r)$.



- **An indicator of clustering objects**

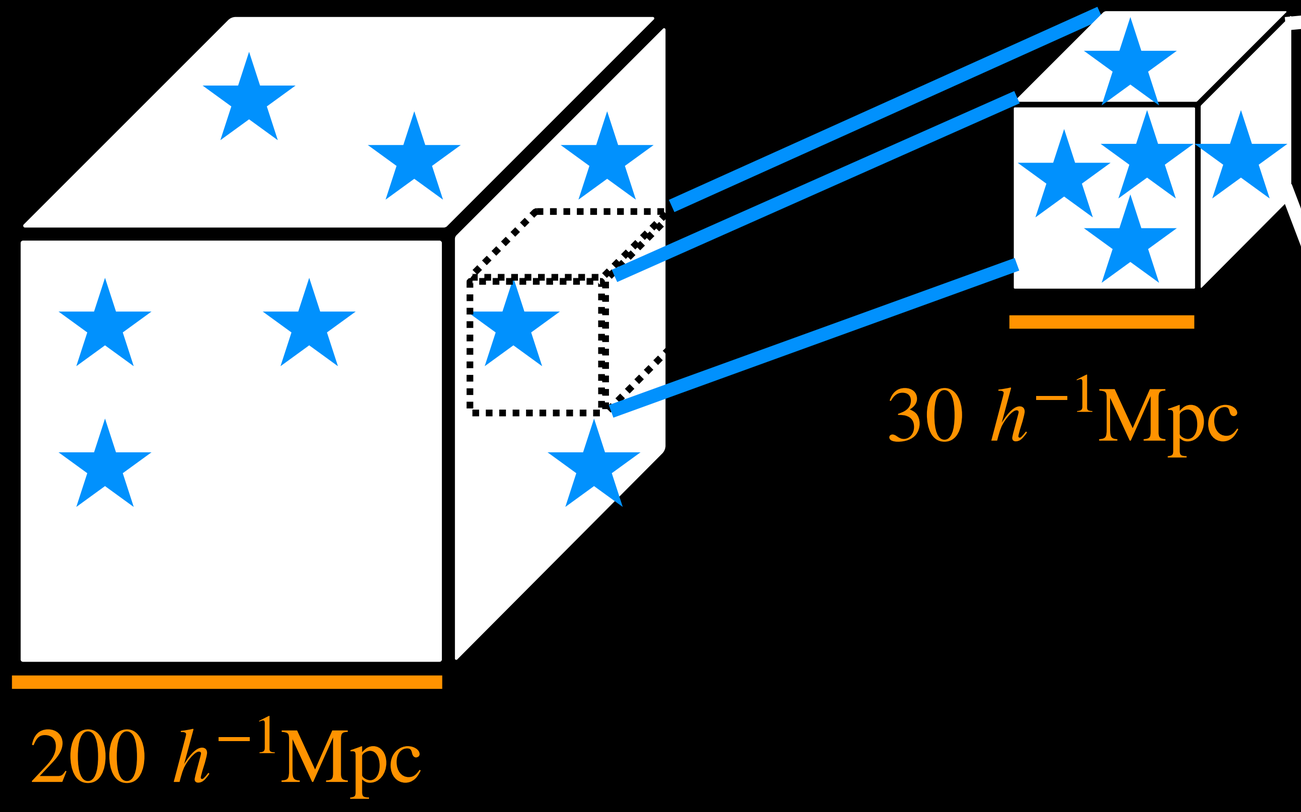
  Large $P_{\mathrm{NN}}(<r)$ in small r —> Strong Clustering!

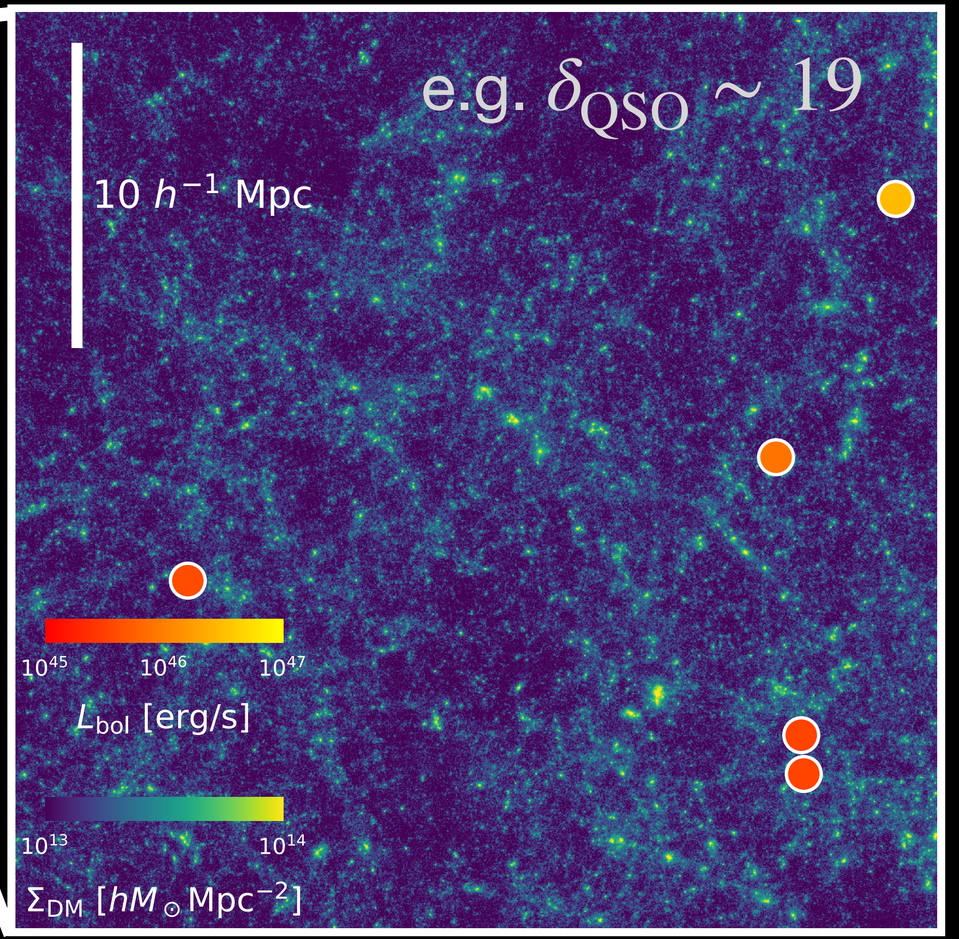
- Sensitive in **small scale clustering** [c.f. $\xi(r)$]
- Include information of higher-order correlation function

Setup: NND for CROCODILE highest overdensities

The whole box

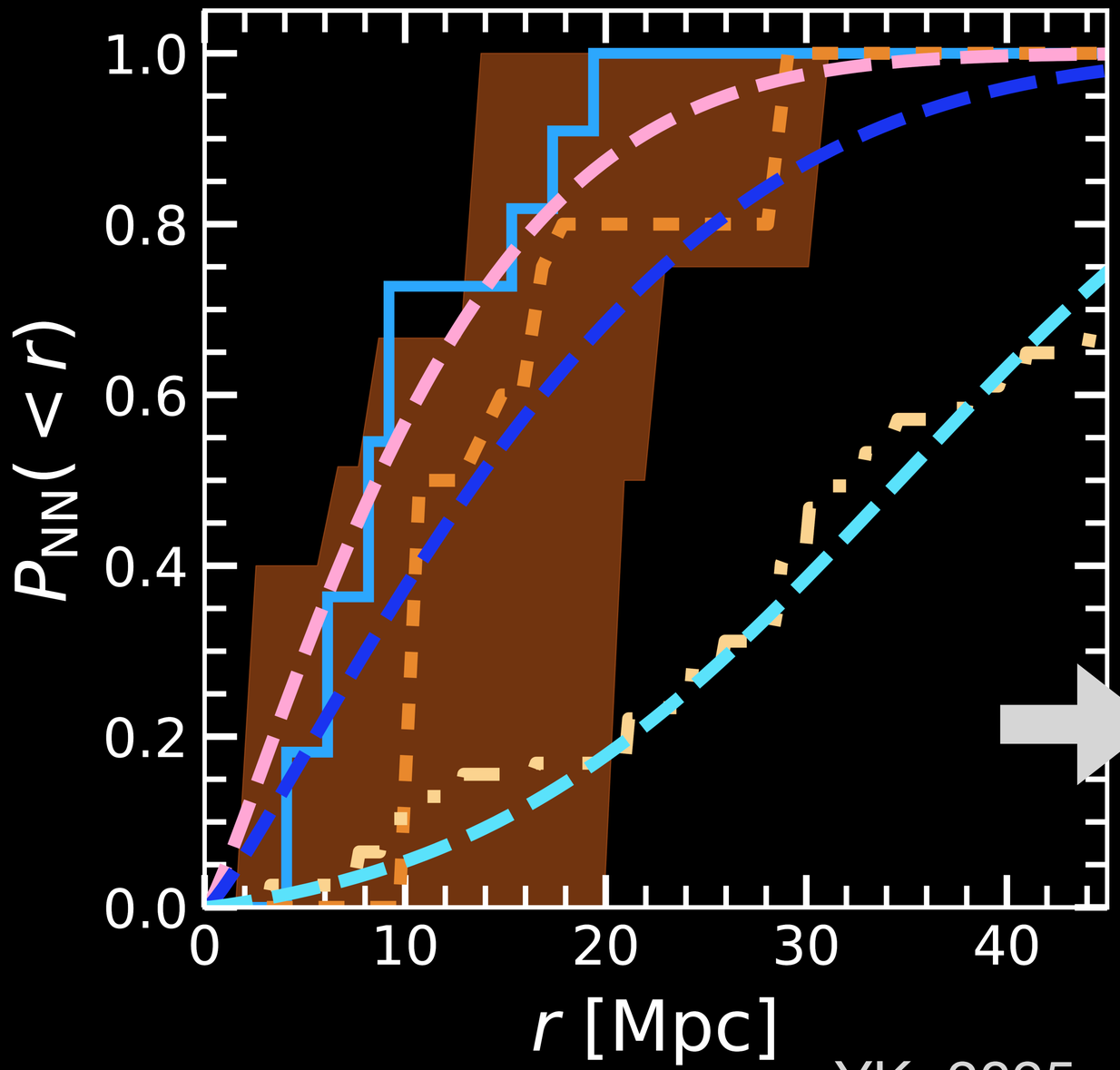
The highest density box of quasars





* 100 independent Type I full-box mock realisations

NND result: Shows Different Clustering?

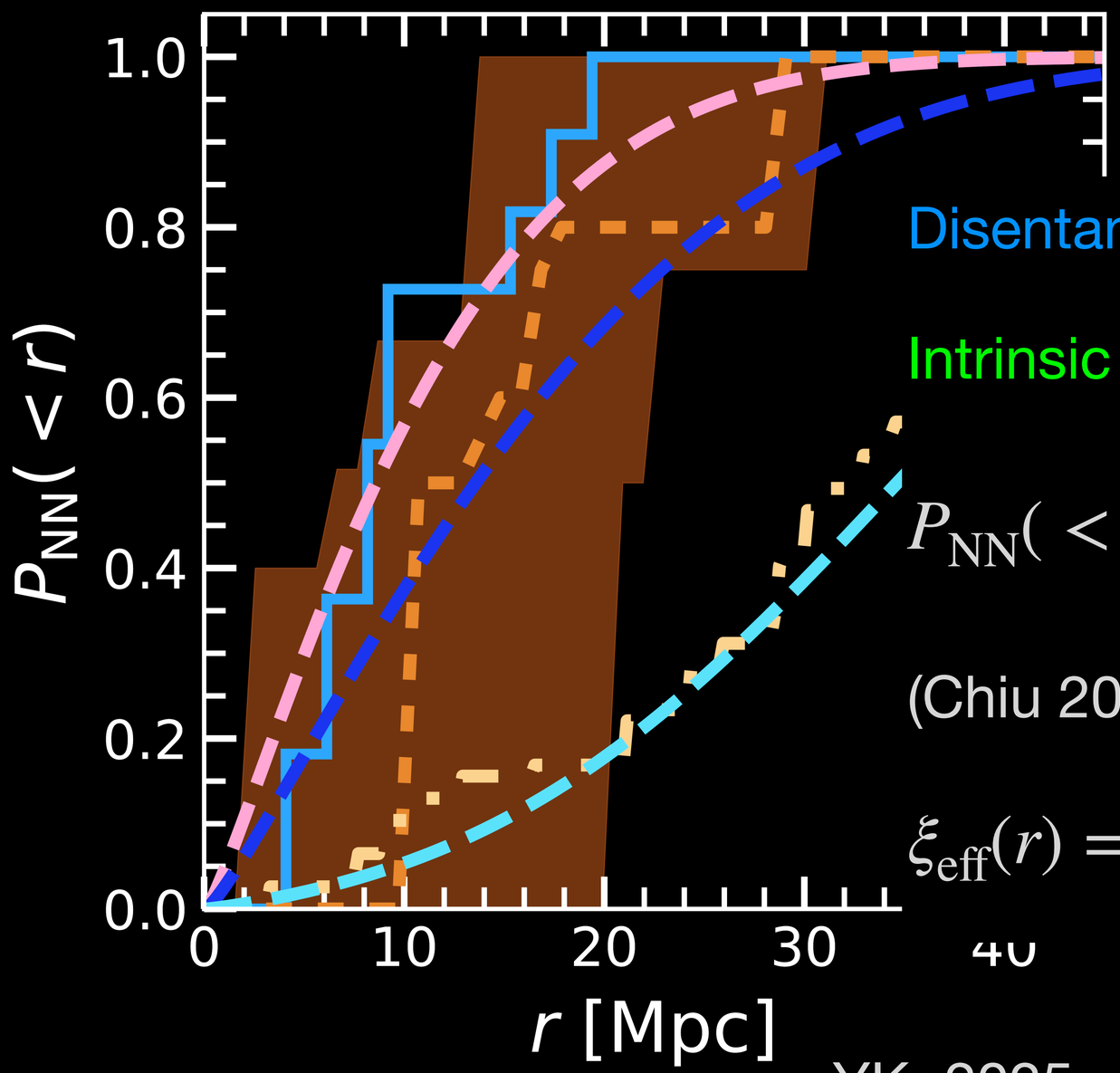


Apparent differences in clustering properties

However, NND conflates mean number density with intrinsic clustering effects.

YK+2025

NND result: Disentangle Intrinsic Clustering



Cosmic Himalayas

Median

Disentangle Clustering from mean density effect $\bar{n}$

Intrinsic Clustering: $\xi_{\rm eff}(r)$

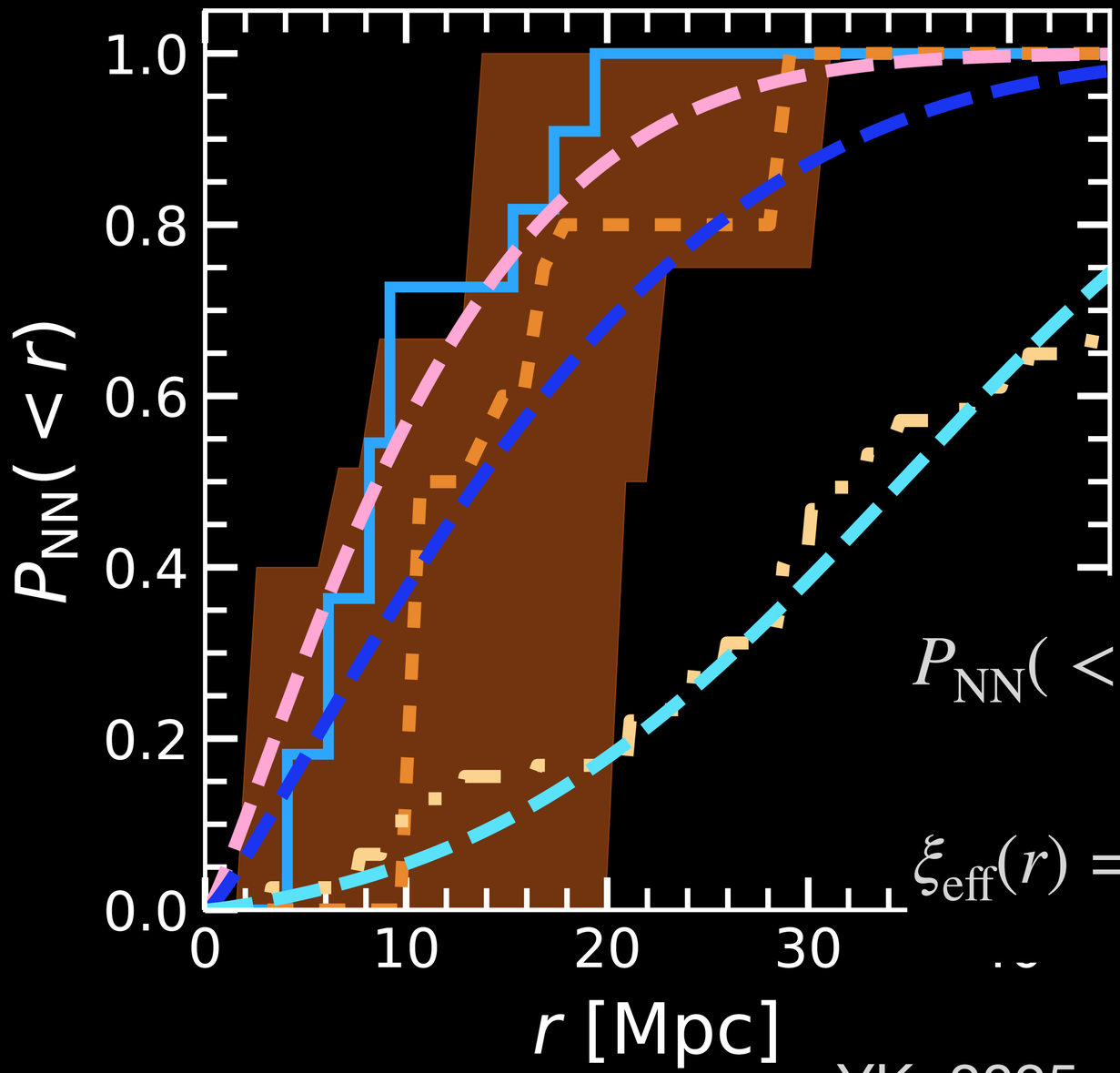
$$P_{\rm NN}(<r) = 1 - \exp\left(- \int_0^r 4\pi r^2 \bar{n} [1+\xi_{\rm eff}(r)] {\rm d}r \right)$$

(Chiu 2013, textbook)

$$\xi_{\rm eff}(r) = \left(\frac{r}{r_0^{\rm eff}} \right)^{-1.8}$$

YK+2025

NND result: Disentangle Intrinsic Clustering



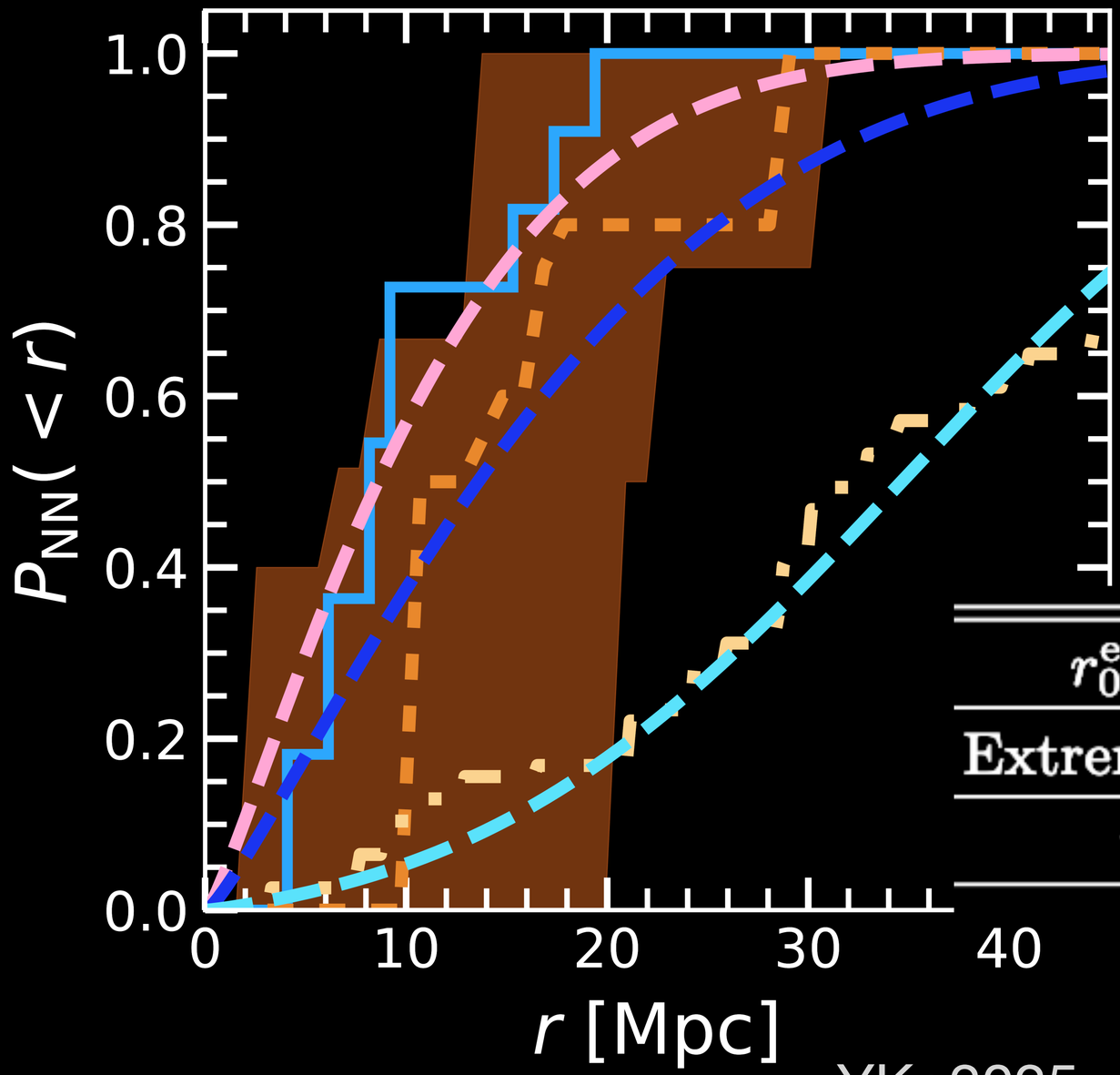
Cosmic Himalayas
Median
1σ scatters
Fit for CH
Fit for Sim
Whole box
Fit for whole box

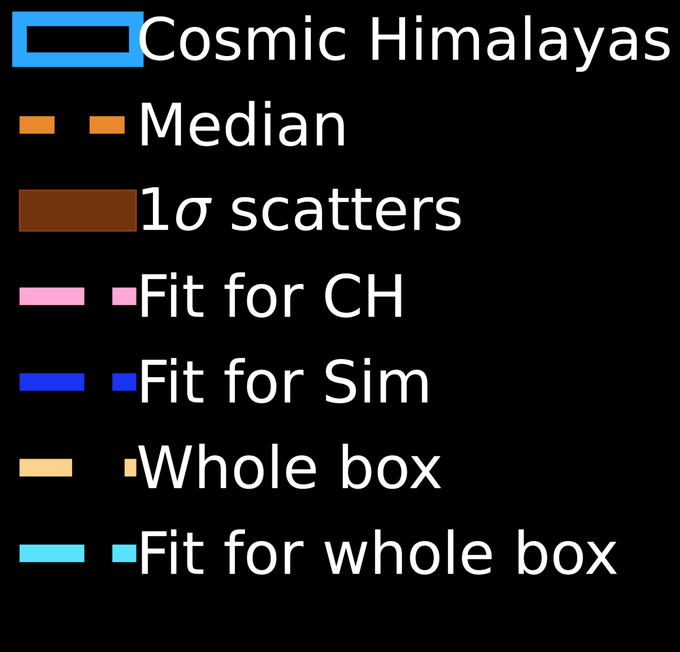
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$$\xi_{\rm eff}(r) = \left(\frac{r}{r_0^{\rm eff}} \right)^{-1.8}$$

YK+2025

NND result: CROCODILE has Similar Intrinsic Clustering





$r_0^{\rm eff}$ $[h^{-1}\,{\rm Mpc}]$	CH	CROCODILE
Extreme overdensities	29.4 ± 0.42	30.9 ± 0.56
whole box	–	8.4 ± 0.35

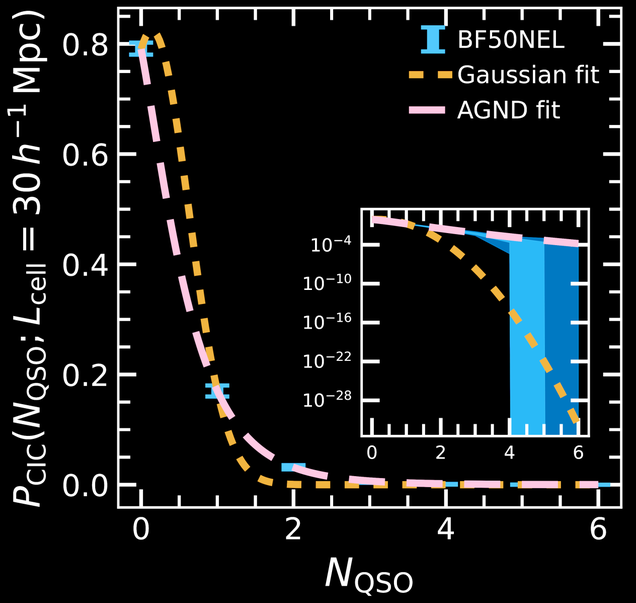
Similar Clustering!

YK+2025

Summary

CIC —> $P_{\rm CIC}(N_{\rm QSO}) \neq$ Gaussian

$P_{\rm CIC}(N_{\rm QSO}) =$ **AGND**

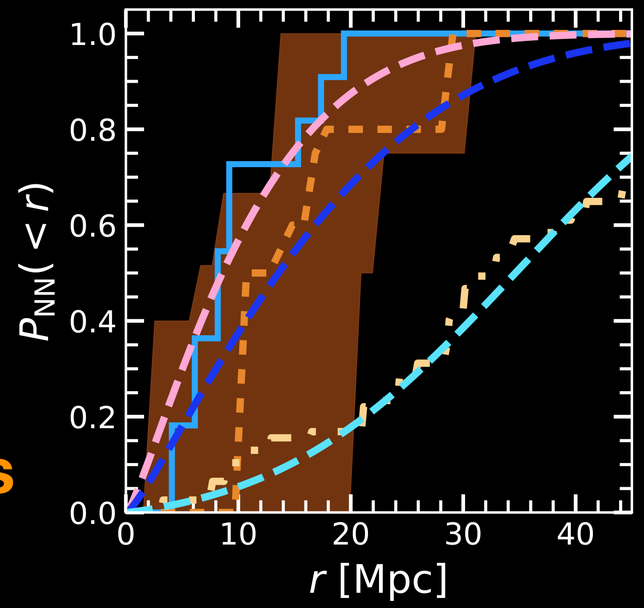


NND —> CROCODILE reproduce **similar clustering properties with CH**

$$r_{0,\rm CH}^{\rm eff} \simeq r_{0,\rm CROC}^{\rm eff} \simeq 30\,h^{-1}\,{\rm Mpc}$$

Cosmic Himalayas ≠ Anomaly overdense region ($\delta = 17\sigma$)

= One of the regions in **non-Gaussian tails**

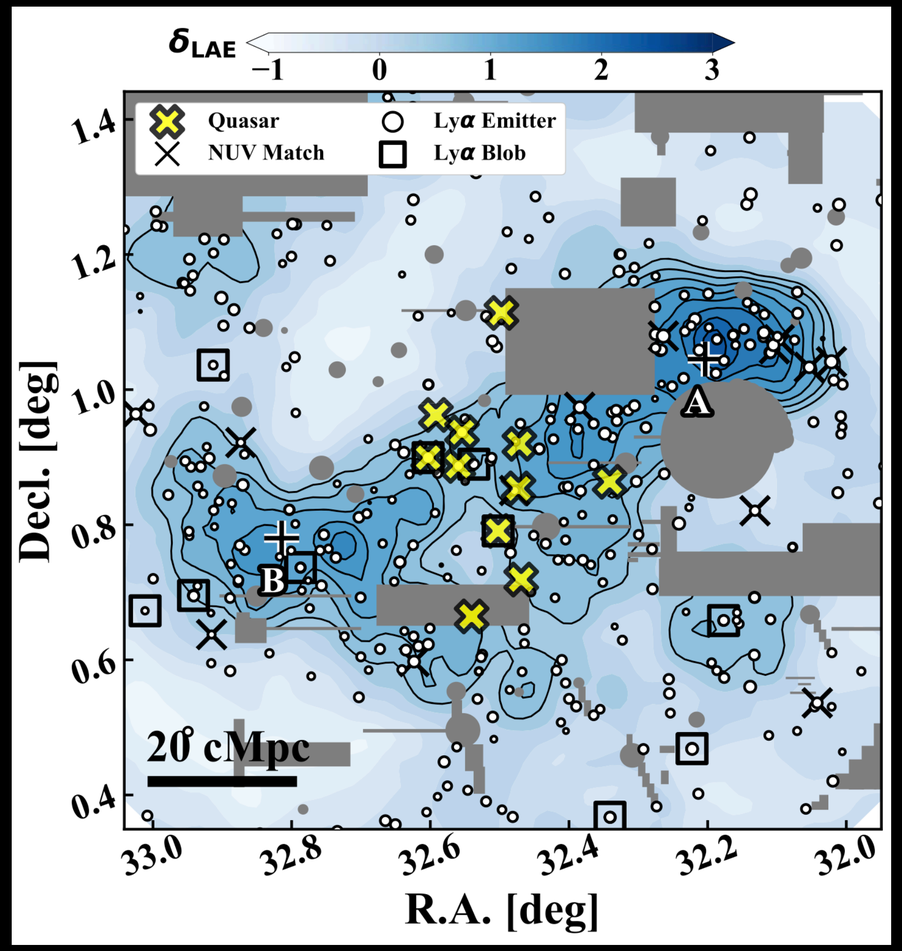


Future Plan: Remaining Open Questions in CH

CH clustering cause the asymmetric distribution of objects and ionisation field

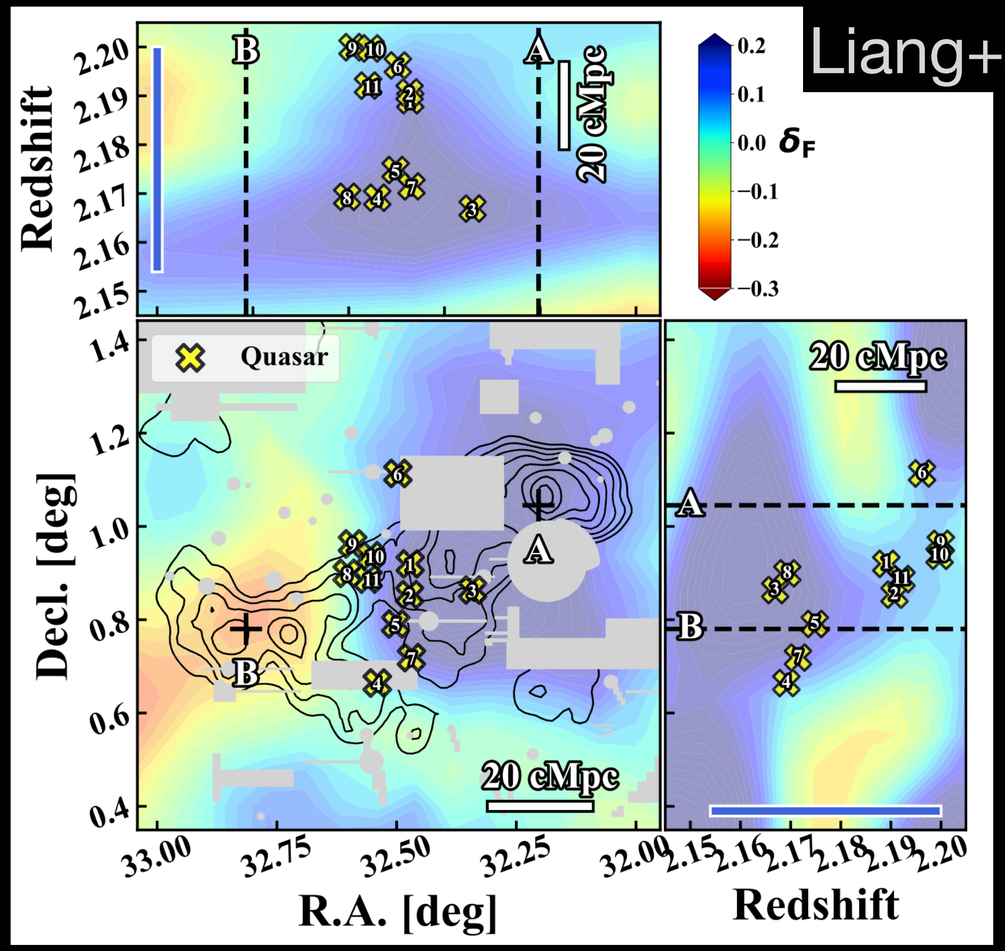
(MOMMOTH-Subaru)

Lyα Emitters dist.

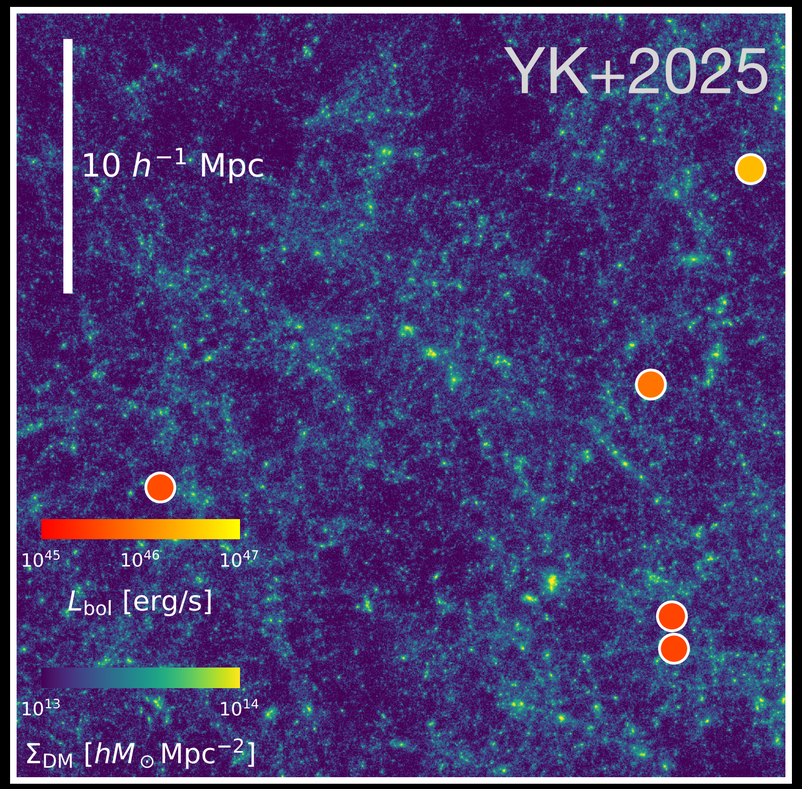


Ionisation field

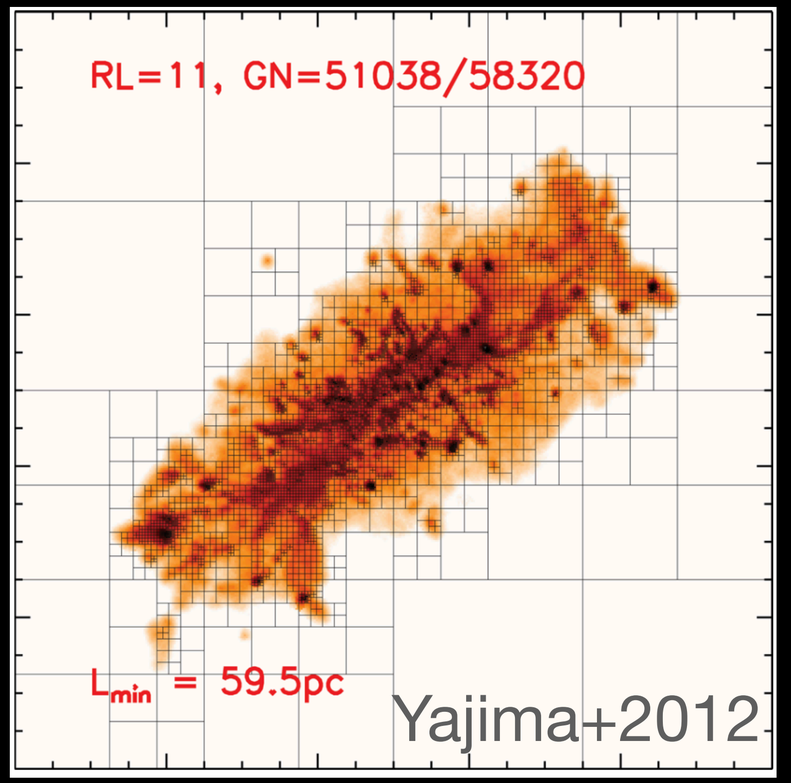
Liang+2025



Future Plan: Radiative Transfer for CROCODILE extreme overdensity



+



Extreme Quasar overdensities

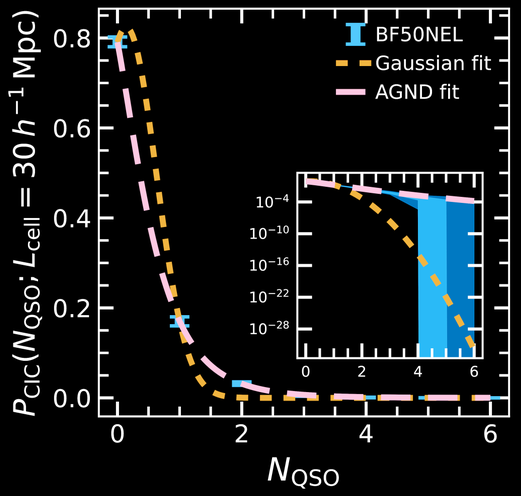
(+Similar clustering) in CROCODILE

Radiative Transfer Code

ART^2 (Li+2008)

Good Laboratory for understanding interaction b/w objects and surrounding media.
(Liang+2025, c.f. Mukae+2020, Daikuhara+2025, Funaki+in prep.)

Summary



arXiv: 2512.24966

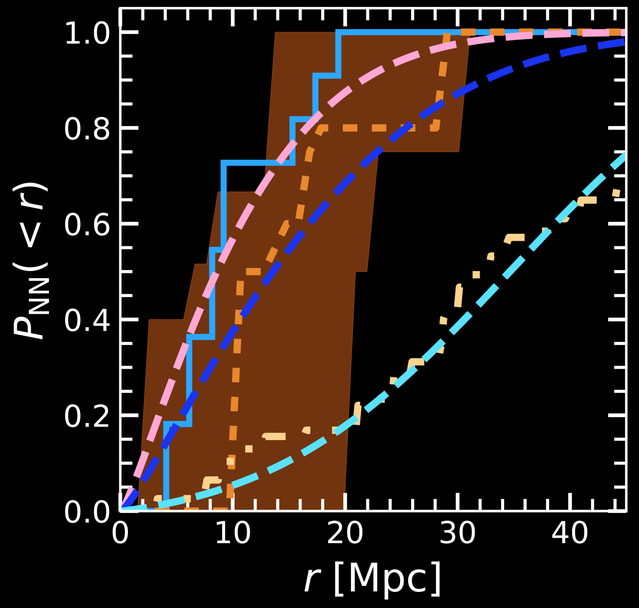
CIC —> $P_{\rm CIC}(N_{\rm QSO}) \neq$ Gaussian

$P_{\rm CIC}(N_{\rm QSO}) =$ **AGND**

NND —> CROCODILE reproduce **similar clustering properties with CH**

YK+2025

$$r_{0,\rm CH}^{\rm eff} \simeq r_{0,\rm CROC}^{\rm eff} \simeq 30\,h^{-1}\,{\rm Mpc}$$



Cosmic Himalayas $\neq$ Anomaly overdense region ($\delta = 17\sigma$)

$=$ One of the regions in **non-Gaussian tails**

Future Plan:

CROCODILE + RT —> understand the interaction b/w objects and surrounding matters.

Appendices

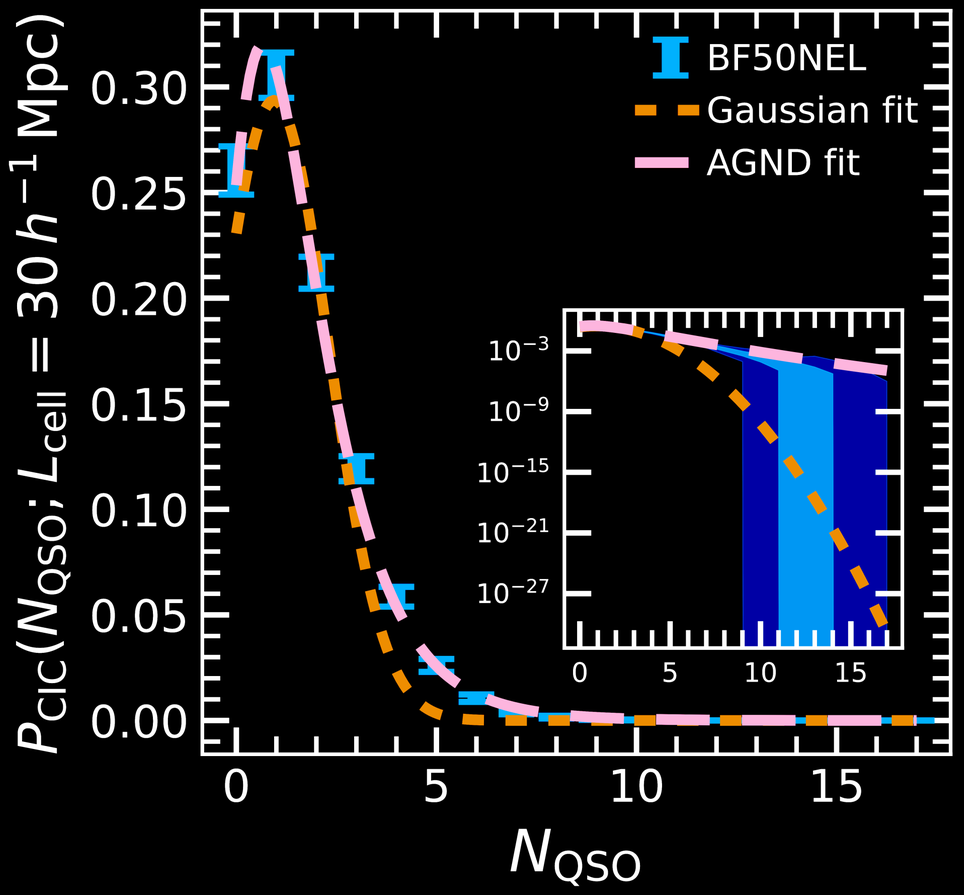
CROCODILE Parameter Set

Parameters	Unit	L200N1024 BoostFactor50 No Eddington Limit (BF50NEL)
Boxsize	$(h^{-1}\mathrm{cMpc})^3$	200^3
Target Redshift of this work	-	2.2
Softening length	$h^{-1}\,\mathrm{ckpc}$	6.76
Maximum softening length	$h^{-1}\,\mathrm{pkpc}$	1.00
DM mass	$h^{-1}M_\odot$	5.39×10^8
Gas mass	$h^{-1}M_\odot$	1.01×10^8
Seed BH mass	$h^{-1}M_\odot$	1×10^5
C_{visc}	-	200π
Boost Factor	-	50
Calculation of BH accretion	-	Angular-momentum-limited Bondi accretion w/o the Eddington accretion limit

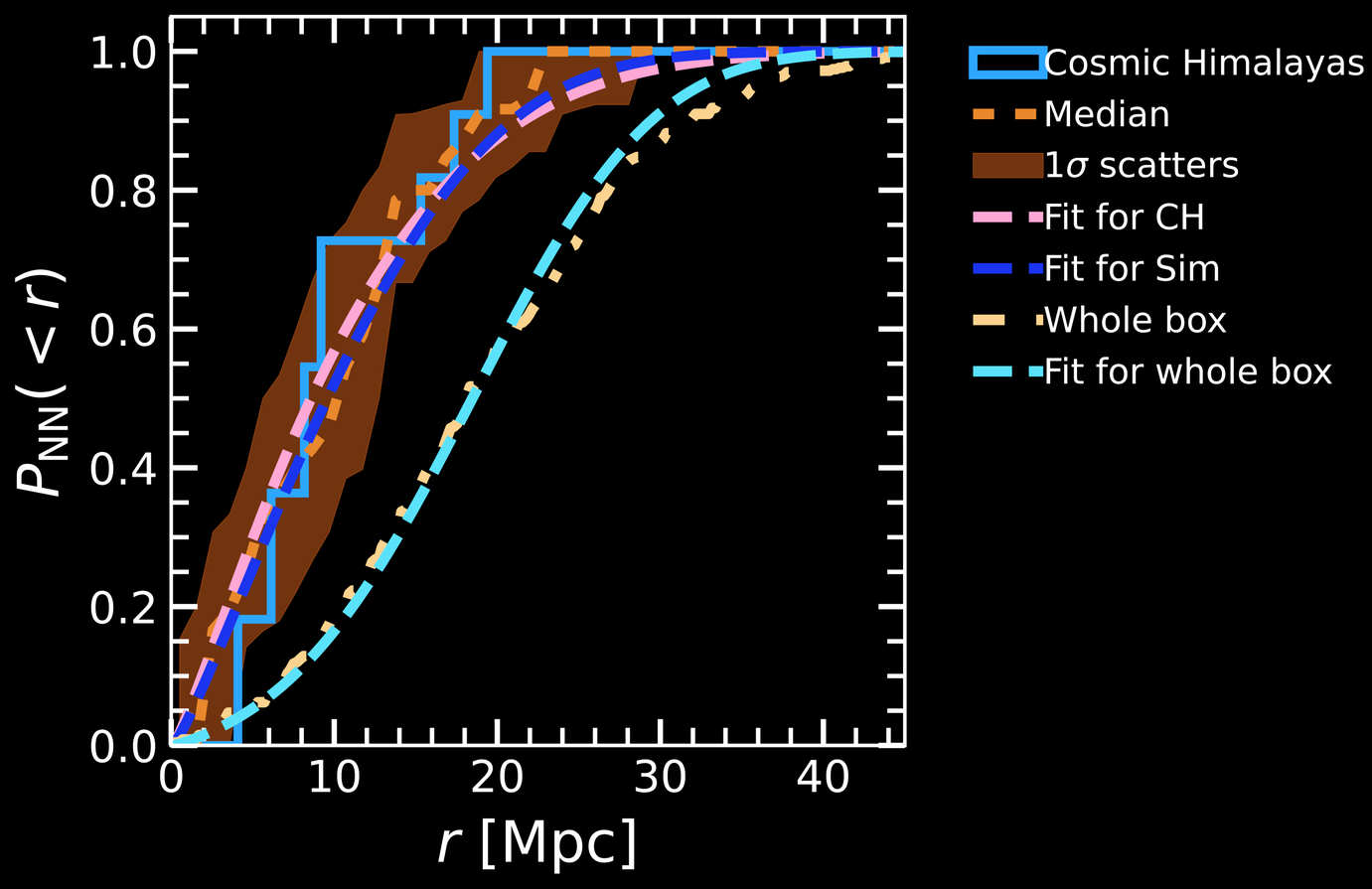
Other Quasar Selection

$$7.1 \leq \log M_{\rm BH}/M_\odot \leq 9.9, \log L_{\rm bol}/{\rm erg\,s^{-1}} > 45$$

CIC



NND



$P_{\rm CIC}$ = AGND ≠ Gaussian

$r_{\rm eff,CROC} \sim 15\,h^{-1}{\rm Mpc}$